Response to the Referee

Dear Editor,

We construct an anomaly-free and renormalizable axial $U_A(1)_{L_{\mu-\tau}}$ model in which a singlet scalar and heavy vector-like leptons generate the charged-lepton masses through a universal-seesaw mechanism. We focus on heavy vector-like leptons, small light–heavy mixing, and $m_s \gtrsim 10\,\mathrm{GeV}$. In this benchmark, the observables considered here depend mainly on $(m_{A'}, g_X)$, and the light charged-lepton current remains dominantly axial, with a small vector component induced by finite right-handed mixing. We confront this benchmark with current experimental searches and examine complementary probes of the chiral interaction structure at a future muon collider.

The referee summarizes our work as follows:

> *The authors construct an anomaly-free, renormalizable model in which an additional Abelian gauge field A' has chiral couplings to muons and taus. This model realizes a limit where the A' coupling to the $\mu - \tau$ current is axial. The second goal of the paper is to examine the phenomenology of this model in the axial limit.*
>
> *The paper presents results that are useful for the model building community. However, I have some significant reservations about key points that either should be included or otherwise justified in being excluded from the manuscript. These are explained below. Revisions for these major points (even if only to clarify why they are not issues) are required before the paper is at the level of a PRD manuscript.*

We sincerely thank the referee for the positive assessment of the model-building construction and for identifying issues that require further clarification or quantitative analysis. We respond to each point below.

Major Points 1).

> *Phenomenology of the S scalar. The second half of the paper reviews a set of experimental bounds on only an effective theory of the A' axially coupled to the $L_{\mu-\tau}$ current. This leaves a major gap between these two parts of the paper: the authors show that their*

UV theory requires a scalar S that cannot be arbitrarily decoupled from the low-energy dynamics, e.g. from Eq. (34) which states that $m_S^2 < 4\pi \left(m_{A'}/(2g_X)\right)^2$.

Reply. The original one-VEV S + VLL construction contains a physical radial scalar s. At tree level,

$$m_{A'} = 2g_X v_s, \qquad m_s^2 = 2\lambda_S v_s^2, \qquad y_{s\ell\ell} \simeq \frac{m_\ell}{v_s}, \quad \ell = \mu, \tau.$$

Thus, the scalar mass and its leading charged-lepton couplings are tied to the UV symmetry-breaking scale and scalar self-coupling; s is not an arbitrary spectator that can simply be removed from the UV theory. The detailed scalar searches and complementary low-mass constraints are addressed under Major Point 1.1, the point-dependent scalar-mass range and perturbativity condition under Major Point 1.2, and the Higgs-portal mixing assumption under Major Point 1.3.

Major Points 1.1).

It is thus an odd choice to exclude the scalar from the phenomenology of the effective theory: if the S is unavoidable at phenomenologically relevant scales, shouldn't one of the main points of the paper be that the only known UV completion of an axial $L_{\mu-\tau}$ gauge boson has a prediction for a scalar particle in a mass range that can be probed by colliders?

Reply. Yes. The radial scalar is a physical and potentially testable prediction of the UV completion. Its collider phenomenology must be assessed using scalar-specific production and decay hypotheses, with the UV branching fractions included when translating published limits.

Direct collider searches and UV mapping.

BaBar and Belle searched for

$$e^+e^- \to \tau^+\tau^- s, \qquad s \to e^+e^-,\, \mu^+\mu^-,$$

over approximately $0.04\,\mathrm{GeV} \lesssim m_s \lesssim 6.5$–$7\,\mathrm{GeV}$ [1, 2]. In the mass-proportional benchmark used in these analyses,

$$\mathcal{L} \supset -\xi \sum_\ell \frac{m_\ell}{v_h} s\bar{\ell}\ell, \qquad \xi = \frac{v_h}{v_s}.$$

The Belle result $\xi^{\rm lim} = 0.483$ at $m_s = 1\,\text{GeV}$ [2] therefore gives

$$v_s > \frac{v_h}{\xi^{\rm lim}} \simeq 509\,\text{GeV}, \qquad y_{s\mu\mu} < 2.1 \times 10^{-4},$$

and, using $m_{A'} = 2g_X v_s$,

$$g_X < 9.8 \times 10^{-4} \left(\frac{m_{A'}}{\text{GeV}}\right)$$

at 90% confidence level for this illustrative $m_s = 1\,\text{GeV}$ point with $m_{A'} > m_s/2$, so that the two-body channel $s \to A'A'$ is closed. This example illustrates the mass-dependent scalar-search translation; it is separate from the $m_s \gtrsim 10\,\text{GeV}$ scalar-suppressed reference benchmark used for Fig. 2. If $m_{A'} < m_s/2$, the $A'A'$ mode is open and the visible-channel limit must be interpreted with the UV branching fraction rather than applied as the same g_X bound. For the μ- and τ-lepton couplings relevant to the associated production and dilepton/ditau channels, the mass-proportional benchmark has the same Yukawa normalization as the radial scalar in our construction:

$$y_{s\ell\ell} = \frac{m_\ell}{v_s} = \frac{2g_X m_\ell}{m_{A'}} = \xi\frac{m_\ell}{v_h}, \qquad \xi = \frac{v_h}{v_s}, \qquad \ell = \mu, \tau.$$

At fixed $(m_{A'}, g_X)$, this Yukawa coupling is independent of m_s; changing m_s changes the scalar self-coupling and the kinematically open decay channels, not the mass-proportional Yukawa normalization. In the minimal benchmark, the radial scalar has no direct tree-level Yukawa coupling to electrons. The electron coupling induced by Higgs–scalar mixing, together with loop corrections to it, is neglected in this comparison. We therefore use the $s \to \mu^+\mu^-$ channel for the $m_s = 1\,\text{GeV}$ Belle comparison and do not directly apply the experimental $s \to e^+e^-$ limits to this benchmark.

For $m_s > 2m_{A'}$, the gauge completion predicts the two-vector partial width

$$\Gamma(s \to A'A') = \frac{m_s^3}{32\pi v_s^2}\left(1 - 4x + 12x^2\right)\sqrt{1-4x}, \qquad x = \frac{m_{A'}^2}{m_s^2},$$

while each kinematically open charged-lepton channel has

$$\Gamma(s \to \ell^+\ell^-) = \frac{m_\ell^2 m_s}{8\pi v_s^2}\left(1 - \frac{4m_\ell^2}{m_s^2}\right)^{3/2}, \qquad \ell = \mu, \tau.$$

When both channels are open, their ratio is

$$\frac{\Gamma(s \to A'A')}{\Gamma(s \to \ell^+\ell^-)} = \frac{m_s^2}{4m_\ell^2}\frac{(1 - 4x + 12x^2)\sqrt{1-4x}}{(1 - 4m_\ell^2/m_s^2)^{3/2}}.$$

Away from the $A'A'$ threshold this ratio can be large, so the $A'A'$ mode can dominate over an individual charged-lepton partial width. The visible branching fraction is consequently

$$\begin{aligned}\mathrm{BR}(s \to \ell^+\ell^-) &= \frac{\Gamma(s \to \ell^+\ell^-)}{\Gamma_s^{\mathrm{tot}}},\\ \Gamma_s^{\mathrm{tot}} &\supset \Theta(m_s - 2m_\mu)\Gamma(s \to \mu^+\mu^-)\\ &\quad + \Theta(m_s - 2m_\tau)\Gamma(s \to \tau^+\tau^-)\\ &\quad + \Theta(m_s - 2m_{A'})\Gamma(s \to A'A').\end{aligned}$$

Thus the event yield in a dilepton search depends on the production rate, this model-dependent visible branching fraction, and the corresponding selection efficiency. When $s \to A'A'$ is open, the resulting A' decays can also change the final-state topology. The published B-factory limits therefore constrain their stated mass-proportional visible-decay benchmark. Their direct translation to the present UV parameter plane applies where the UV branching fractions reproduce the assumed visible benchmark; when $s \to A'A'$ is open, the interpretation instead requires the model-dependent visible branching fractions and the corresponding final-state selection.

Belle II scalar-specific searches.

Belle II has performed complementary searches for scalar signals in the $\mu^+\mu^-\tau^+\tau^-$ and four-muon final states. The $e^+e^- \to \mu^+\mu^- X$, $X \to \tau^+\tau^-$ analysis covers 3.6–10 GeV and tests a Higgs-like spin-zero leptophilic scalar with charged-lepton couplings proportional to their masses. For the μ and τ couplings relevant to this process, that benchmark has the same Yukawa normalization as the radial scalar in our one-VEV construction [3]. The four-muon search, $e^+e^- \to \mu^+\mu^- X$ with $X \to \mu^+\mu^-$, covers 0.212–9 GeV and includes a muonphilic-scalar hypothesis alongside the $L_\mu - L_\tau$ vector boson [4].

In our model, the radial scalar couples to both μ and τ. Once $s \to \tau^+\tau^-$ is open, this additional decay channel contributes to the total width and affects the dimuon branching fraction relevant to the four-muon search. The channel $s \to A'A'$, when open, also modifies the visible branching fractions and can produce cascade final states with different selection efficiencies. The Belle II results therefore provide complementary scalar-search information; their translation to our UV parameters uses the model-specific production rate, branching fractions, and corresponding event selection. General singlet–Higgs mixing and additional-scalar searches provide further complementary context [5, 6, 7].

Complementary low-mass constraints and fifth-force matching.

The low-mass scalar constraints relevant to Table II are summarized below. In the original one-VEV construction, the scalar–muon Yukawa coupling is related to the UV parameters by

$$y_{s\mu\mu} = \frac{2 g_X m_\mu}{m_{A'}}.$$

The entries probe different observables. In particular, fifth-force and photon-mediated gamma-ray interpretations involve loop-induced couplings and the source-specific assumptions discussed below.

Summary of selected low-mass constraints.

Scalar-mass range	Published constraint	Source-reported bound / applicability
$10^{-17}\,\mathrm{eV} \lesssim m_s \lesssim 2\times10^{-15}\,\mathrm{eV}$	Binary-pulsar periastron advance (95% CL), ultralight scalar [8]	$\mathcal{O}(10^{-21})$ at the best-sensitivity point $m_s \simeq 4\times10^{-16}\,\mathrm{eV}$; not a uniform bound over the interval
$m_s \lesssim 30\,\mathrm{meV}$ (crossover scale, not a hard endpoint)	Fifth-force tests from the muon-only loop-induced scalar–proton coupling [9, 10, 11, 12]	Mass-dependent source curve; ~ 30 meV is the crossover below which it overtakes astrophysical limits; no universal direct $y_{s\mu\mu}$ limit
$m_s \lesssim 100\,\mathrm{keV}$	Neutron-star cooling [9]	$\mathcal{O}(10^{-13})$; Table 1 gives 7.4×10^{-14} for $m_s \lesssim 1\,\mathrm{keV}$
$m_s \lesssim \mathcal{O}(10)\,\mathrm{MeV}$	SN 1987A neutrino energy loss, conservative cold-SFHo free-streaming benchmark [13, 9]	$\lesssim 4.2\times10^{-9}$ over the mass range shown in Caputo et al.'s Fig. 2
$m_s \lesssim \mathcal{O}(10)\,\mathrm{MeV}$	SN 1987A γ-ray fluence, cold-SN benchmark [13]	$g_{\phi\mu}^{\mathrm{source}} < 1.11\times10^{-10}\sqrt{\mathrm{MeV}/m_s}$; mass dependent and stronger than neutrino cooling for $m_s \gtrsim 0.70\,\mathrm{keV}$
$0.1\,\mathrm{MeV} \lesssim m_s \lesssim \mathcal{O}(10)\,\mathrm{MeV}$	Diffuse supernova γ-ray background, short-lived scalar branch [13]	$g_{\phi\mu}^{\mathrm{source}} \lesssim 4.2\times10^{-11}$ for $n_7^{\mathrm{cc}} = 0.58$; the long-lived branch is mass dependent

The entries refer to distinct observables and source assumptions and are not combined into a single exclusion band.

Binary-pulsar constraint.

The binary-pulsar entry uses Liu and Tang's scalar-mediator result, whose coupling has the same Yukawa normalization as $y_{s\mu\mu}$. We quote the $\mathcal{O}(10^{-21})$ best-sensitivity point separately from the reported mass interval; it is not

a uniform limit across that interval. The $L_\mu - L_\tau$ vector-mediator result of Dror et al. is a distinct constraint [8, 14].

Neutron-star and supernova neutrino cooling.

Fiorillo et al.'s v2 neutron-star cooling result is mass dependent up to approximately 100 keV; the more specific 7.4×10^{-14} value in their Table 1 applies only for $m_s \lesssim 1\,\mathrm{keV}$ and must not be read as a constant bound over the full mass interval [9]. For SN 1987A neutrino energy loss, Fiorillo et al.'s Table 1 quotes 4.2×10^{-9} for its $m_s \lesssim 1\,\mathrm{keV}$ benchmark, whereas Caputo et al.'s Fig. 2 displays the conservative cold-SFHo free-streaming boundary out to $m_s \sim \mathcal{O}(10)\,\mathrm{MeV}$, remaining approximately at that level over the plotted range. We use the mass coverage of the latter curve, not the mass label of the former tabulated benchmark. This is a source-specific free-streaming cooling criterion, not a complete trapping or explosion-energy exclusion [13, 9].

Photon-mediated supernova signals.

The SN 1987A gamma-ray fluence and diffuse-supernova gamma-ray background are separate observables, listed separately in Table II. The quoted photon-mediated results use Caputo et al.'s source normalization $g_{\phi\mu}^{\mathrm{source}}$ and muon-only treatment; their production and loop approximations are restricted to $m_\phi \lesssim 10\,\mathrm{MeV}$ [13].

Fifth-force matching.

The fifth-force result is an indirect constraint. In Fiorillo et al.'s low-momentum, muon-only benchmark, the muon triangle induces a scalar–photon interaction and an effective proton coupling,

$$g_{\phi p}^{(\mu)} = \frac{\alpha^2}{4} g_{\phi\mu}.$$

Ordinary-matter fifth-force experiments constrain this loop-induced proton coupling, rather than a tree-level scalar–muon Yukawa vertex. The mass-dependent curve in Fig. 1 of Fiorillo et al. overtakes the astrophysical limits below an approximate crossover scale of $m_\phi \sim 30\,\mathrm{meV}$; this is not a sharp upper endpoint of experimental sensitivity [9].

In the one-VEV UV construction, the radial scalar couples to both charged leptons,

$$y_{s\mu\mu} = \frac{m_\mu}{v_s}, \qquad y_{s\tau\tau} = \frac{m_\tau}{v_s}, \qquad \frac{y_{s\mu\mu}}{m_\mu} = \frac{y_{s\tau\tau}}{m_\tau} = \frac{1}{v_s}.$$

Consequently, in the fifth-force regime $m_s \lesssim 30\,\mathrm{meV} \ll 2m_\mu$, the low-

momentum electromagnetic form factor receives equal-sign contributions from both loops:

$$\mathcal{F}_{s\gamma\gamma}(0) \propto \sum_{\ell=\mu,\tau} Q_\ell^2 \frac{y_{s\ell\ell}}{m_\ell} = \frac{2}{v_s}.$$

The contributions add because the amplitude is weighted by Q_ℓ^2, so the opposite $U(1)_{L_\mu - L_\tau}$ charges do not cause a cancellation. Fiorillo et al.'s curve is therefore a muon-only source benchmark, not the complete fifth-force prediction of this UV model; translating it requires matching both the muon and tau loops.

UV interpretation and scalar decays.

For application of the source-level limits to the UV scalar, the decay $s \to A'A'$ must be included whenever $m_s > 2m_{A'}$. When open, it contributes to the total width and can change the lifetime, visible branching fractions, cascade topology, and energy deposition relevant to a given search or astrophysical observable. The entries in Table II are therefore source-level constraints; their UV interpretation depends on the corresponding production and decay assumptions.

For a source result that directly constrains the same scalar–muon Yukawa normalization, the one-VEV relation given above yields the conditional translation

$$g_X \lesssim \frac{y_{s\mu\mu}^{\rm lim}(m_s) m_{A'}}{2m_\mu},$$

at a specified m_s within the source's mass range. This is not a model-independent bound on g_X. In particular, the fifth-force curve cannot be inserted into this expression as a direct Yukawa limit; its UV translation first requires the two-lepton loop matching.

Scalar contribution to the muon anomalous magnetic moment.

The scalar contribution to the muon anomalous magnetic moment is included together with the axial-vector contribution,

$$\Delta a_\mu = \Delta a_\mu^{A'} + \Delta a_\mu^s, \qquad \Delta a_\mu^s = \frac{y_{s\mu\mu}^2}{8\pi^2} \int_0^1 dx \, \frac{x^2(2-x)m_\mu^2}{m_s^2(1-x) + x^2 m_\mu^2}.$$

For the reference regime $m_s \gtrsim 10\,\mathrm{GeV}$, the scalar term is below 0.2% of the pure-axial A' contribution. For lighter s, the positive scalar term can instead compete with, or partially cancel, the negative axial contribution. Accordingly,

the $(g-2)_\mu$ contour in Fig. 2 is evaluated for the stated scalar-suppressed benchmark, rather than being applied to arbitrary scalar masses.

The radial scalar therefore has testable phenomenology across different mass ranges. BaBar and Belle probe its production with a tau pair, while Belle II searches for scalar signals in the $\mu^+\mu^-\tau^+\tau^-$ and four-muon final states. At lower masses, binary pulsars, fifth-force experiments, and stellar or supernova observations provide complementary probes. The scalar also contributes to $(g-2)_\mu$, with a mass-dependent impact. The charged-lepton and $A'A'$ partial widths enter the visible branching fractions, while $s \to A'A'$, when open, can produce cascade final states; translating a published visible-channel limit therefore uses the production and decay structure of the one-VEV UV model.

Major Points 1.2).

> *What is the range of phenomenologically allowed S mass including perturbativity limits on λ and a motivated range of $m_{A'}$ and g_X. For example: perhaps the authors imagine a range of $m_{A'}$ below some benchmark mass where certain experimental constraints on g_X become very strong. Then with that set of constraints (which should be stated clearly, see minor point #1 below), one may give a range of m_S and comment on the experimental constraints.*

Reply. In the one-VEV construction, the scalar-mass range follows point by point from the UV relations, rather than defining a fixed interval independently of $(m_{A'}, g_X)$. The relevant relations are

$$v_s = \frac{m_{A'}}{2g_X}, \qquad \lambda_S = \frac{m_s^2}{2v_s^2} = \frac{2g_X^2 m_s^2}{m_{A'}^2}.$$

For the scalar-suppressed reference benchmark $m_s \gtrsim 10\,\mathrm{GeV}$ and the perturbativity criterion $\lambda_S \leq 4\pi$, the corresponding point-dependent range is

$$10\,\mathrm{GeV} \lesssim m_s \leq \sqrt{8\pi}\, v_s = \sqrt{2\pi}\, \frac{m_{A'}}{g_X}.$$

It requires

$$v_s \gtrsim \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}, \qquad m_{A'} \gtrsim (3.99\,\mathrm{GeV})\, g_X.$$

This is the theoretical consistency boundary represented by the $v_s \simeq 2\,\mathrm{GeV}$ line in Fig. 2. At each plotted point, v_s is derived and the scalar self-coupling is checked against the perturbativity criterion.

The lower-mass scalar regimes and the direct BaBar/Belle/Belle II searches are discussed under Major Point 1.1. The point-dependent perturbative mass range for the A' benchmark is distinct from the mass-dependent scalar phenomenology. The dashed line is a theoretical consistency condition for the specified $m_s \gtrsim 10\,\mathrm{GeV}$ benchmark, not an experimental exclusion or a universal boundary for every possible scalar-sector choice.

Major Points 1.3).

> *Above (34): the authors "assume negligible Higgs-portal mixing." Is this interaction not generated at loop level? It seems that S-h mixing cannot be avoided due to μ and τ loops. In turn, this seems like it would lead to strong experimental bounds on the UV completion. In fact, once there is mixing, is there not a resonance where the mixing becomes large as $m_S \to m_h$? It would be useful to quantify the size of this mixing over the allowed range of m_S. Existing Higgs measurements may set the strongest bounds on the UV completion. If these bounds are obviously negligible (though again, I expect this to be strongly constrained as $m_S \to m_h$), then the authors should present this as the justification for ignoring m_S in the low-energy theory. (Though I am still skeptical that the S can be ignored since it cannot formally decouple.)*

Reply. We agree that a vanishing Higgs-portal coupling is not enforced by the gauge symmetry: the operator $(H^\dagger H)(S^\dagger S)$ is allowed and contributes $(M^2_{\mathrm{loc}})_{hs} = \lambda_{HS} v_h v_s$ after symmetry breaking, so the portal boundary condition must be stated at a matching scale and in a specified renormalization convention.

Generic singlet–Higgs mixing is constrained by the 125 GeV Higgs signal strengths and direct searches for additional scalar states; representative analyses are cited above. For the original one-singlet S + VLL UV model considered here, we complement these general results with the model-specific VLL threshold matching and the finite-momentum h-s self-energy, evaluated using the full charged-lepton/VLL mass-eigenstate matrices as detailed below.

In the $\overline{\mathrm{MS}}$ scheme, with $\mathcal{L} \supset -V$ and $V \supset \lambda_{HS}(H^\dagger H)(S^\dagger S)$, this threshold follows from the fermionic Coleman–Weinberg effective potential [15, 16]. For

constant neutral Higgs and singlet backgrounds, the charged-lepton/VLL mass matrix for each flavour is

$$\mathcal{M}_\ell(H,S)=\begin{pmatrix}0 & y_\ell H^0\\ \lambda_\ell S_\ell & M_E\end{pmatrix},\qquad S_\mu=S,\quad S_\tau=S^*,$$

where H^0 is the neutral Higgs background. Defining

$$X_\ell=|y_\ell|^2H^\dagger H,\qquad Z_\ell=|\lambda_\ell|^2S^\dagger S,$$

the two eigenvalues of $\mathcal{M}_\ell^\dagger\mathcal{M}_\ell$ are

$$m^2_{\ell,\pm}=\frac{1}{2}\left[M_E^2+X_\ell+Z_\ell\pm\sqrt{(M_E^2+X_\ell+Z_\ell)^2-4X_\ell Z_\ell}\right].$$

The one-loop contribution of these two Dirac mass eigenstates is

$$V^F_{1,\ell}=-\frac{1}{16\pi^2}\sum_{a=\pm}m^4_{\ell,a}\left[\ln\left(\frac{m^2_{\ell,a}}{\mu_R^2}\right)-\frac{3}{2}\right],$$

where the factor of four Dirac degrees of freedom is included. In the matching expansion $X_\ell, Z_\ell \ll M_E^2$,

$$m^2_{\ell,+}=M_E^2+X_\ell+Z_\ell-\frac{X_\ell Z_\ell}{M_E^2}+\mathcal{O}(M_E^{-4}\Phi^6),\qquad m^2_{\ell,-}=\frac{X_\ell Z_\ell}{M_E^2}+\mathcal{O}(M_E^{-4}\Phi^6),$$

with Φ denoting the background scalar fields. To display the mixed term explicitly, set $f(t)=t^2[\ln(t/\mu_R^2)-3/2]$. The coefficient of $X_\ell Z_\ell$ in the heavy-eigenstate contribution is

$$-\frac{f'(M_E^2)}{M_E^2}X_\ell Z_\ell+f''(M_E^2)X_\ell Z_\ell=2X_\ell Z_\ell,$$

so the logarithms cancel. The light-eigenstate contribution starts at order $X_\ell^2Z_\ell^2/M_E^4$ and does not contribute to the dimension-four portal coefficient. Thus

$$\Delta V^{(1)}_{\rm VLL}\Big|_{(H^\dagger H)(S^\dagger S)}=-\frac{1}{8\pi^2}\sum_{\ell=\mu,\tau}X_\ell Z_\ell=-\frac{1}{8\pi^2}\sum_{\ell=\mu,\tau}|y_\ell|^2|\lambda_\ell|^2(H^\dagger H)(S^\dagger S),$$

which gives the finite one-loop threshold

$$\Delta\lambda^E_{HS}=-\frac{1}{8\pi^2}\sum_{\ell=\mu,\tau}|y_\ell|^2|\lambda_\ell|^2.$$

At this order the mixed quartic coefficient is independent of M_E and μ_R; the M_E-dependence first appears in higher-dimensional terms. This is a zero-external-momentum matching coefficient, not an RG beta-function contribution. At the matching scale $\mu_M = M_E$, the coupling relation is

$$\lambda_{HS}^{\rm EFT}(\mu_M) = \lambda_{HS}^{\rm UV}(\mu_M) + \Delta\lambda_{HS}^{E}.$$

The finite-momentum h-s self-energy and the associated mixing diagnostics are evaluated separately below. We use $\lambda_{HS}^{\rm EFT}(M_E) = 0$ as an explicit small-portal reference condition, which requires a nonzero UV portal $\lambda_{HS}^{\rm UV} = -\Delta\lambda_{HS}^{E}$. This condition is a benchmark choice, not a symmetry prediction. In the leading charged-lepton/VLL mass-mixing approximation,

$$\theta_{\ell L} \simeq \frac{y_\ell v_h}{\sqrt{2}M_E}, \qquad \theta_{\ell R} \simeq \frac{\lambda_\ell v_s}{\sqrt{2}M_E}, \qquad m_\ell \simeq \theta_{\ell L}\theta_{\ell R} M_E \simeq \frac{y_\ell \lambda_\ell v_h v_s}{2M_E}.$$

For illustration, taking $M_E = 20\,\text{TeV}$, $\theta_{\mu L} = \theta_{\tau L} = 10^{-2}$, and the leading charged-lepton/VLL mass-mixing relations gives $\Delta\lambda_{HS}^{E} \simeq -0.2108$ at $(m_{A'}, g_X, v_s) = (10\,\text{GeV}, 0.0705, 70.9\,\text{GeV})$, and $\Delta\lambda_{HS}^{E} \simeq -0.01067$ at $(63\,\text{GeV}, 0.1, 315\,\text{GeV})$. For these illustrative points only, we additionally impose $|\lambda_\ell| \le \sqrt{4\pi}$; this is a theory-consistency benchmark rather than a new exclusion contour in Fig. 2. The comparison demonstrates that setting the UV portal to zero is not equivalent to setting the matched EFT portal to zero, and that the latter must be stated as an input if the small-portal benchmark is retained.

The condition $\lambda_{HS}^{\rm EFT}(M_E) = 0$ removes the local dimension-four mixed mass insertion from the matched EFT at the matching scale; it does not set the UV portal to zero. In particular, $\lambda_{HS}^{\rm UV}(M_E) = -\Delta\lambda_{HS}^{E}$ is nonzero in both benchmarks. The CP-even radial-scalar bubbles generated by this portal can be organized using

$$g_{hhh} = 6\lambda_H v_h, \qquad g_{sss} = 6\lambda_S v_s, \qquad g_{hhs} = \lambda_{HS}^{\rm UV} v_s, \qquad g_{hss} = \lambda_{HS}^{\rm UV} v_h.$$

Their zero-momentum-subtracted contribution has the form

$$\begin{aligned}\Delta\Pi_{hs}^{(h,s)\,\text{subset}} = \frac{\lambda_{HS}^{\rm UV} v_h v_s}{32\pi^2}\Big[&6\lambda_H \Delta\bar{B}_{hh} + 6\lambda_S \Delta\bar{B}_{ss} \\ &+ 2\lambda_{HS}^{\rm UV}\Delta\bar{B}_{hs}\Big], \\ \Delta\bar{B}_{ij} \equiv \Delta\bar{B}_0(m_h^2; m_i^2, m_j^2).&\end{aligned}$$

where the finite, subtraction-scale-independent function is

$$\Delta\bar{B}_0(s; a, b) = -\int_0^1 dx\, \ln\left[1 - \frac{x(1-x)s}{xa + (1-x)b}\right], \qquad a = m_i^2, \quad b = m_j^2.$$

For equal internal masses, this integral has the elementary closed form

$$\boxed{\Delta\bar{B}_0(s;a,a) = 2 - 2\sqrt{\frac{4a}{s}-1}\,\arctan\left(\frac{1}{\sqrt{4a/s-1}}\right), \qquad 0 < s < 4a.}$$

For unequal internal masses, define

$$\begin{aligned}\Lambda(s,a,b) &= (s-a-b)^2 - 4ab,\\ q &= \frac{\sqrt{-\Lambda(s,a,b)}}{2s},\\ t_0 &= \frac{a-b-s}{2s}, \qquad t_1 = t_0 + 1.\end{aligned}$$

and

$$\begin{aligned}G(t) &= t\ln\left[\frac{s(t^2+q^2)}{\mu_0^2}\right] - 2t + 2q\arctan\left(\frac{t}{q}\right),\\ I_0(a,b;\mu_0) &= \frac{a\ln(a/\mu_0^2) - b\ln(b/\mu_0^2)}{a-b} - 1.\end{aligned}$$

The mixed-mass result is then

$$\boxed{\Delta\bar{B}_0(s;a,b) = I_0(a,b;\mu_0) - \big[G(t_1) - G(t_0)\big], \qquad a \neq b, \quad \Lambda(s,a,b) < 0.}$$

The arbitrary reference scale μ_0 cancels from this difference. All benchmark evaluations below lie on this real sub-threshold branch. Thus the hh and ss terms use the equal-mass expression, while the hs term uses the mixed-mass expression. At $m_s = 100\,\mathrm{GeV}$, the function triplet is $(\Delta\bar{B}_{hh}, \Delta\bar{B}_{ss}, \Delta\bar{B}_{hs}) = (0.1862006, 0.3150601, 0.2366921)$; at $m_s = m_h$, all three equal 0.1862006. The closed forms agree with independent 8,000-interval Simpson integration at relative differences below 10^{-14}. Because $\lambda_{HS}^{\mathrm{UV}}$ is fixed at the one-loop VLL-threshold order, the terms with one portal insertion are next order in this combined counting; the term with two portal insertions is one order further. In an R_ξ gauge, the corresponding Goldstone vertices are $g_{hG_HG_H} = 2\lambda_H v_h$, $g_{sG_HG_H} = \lambda_{HS}^{\mathrm{UV}} v_s$, $g_{haa} = \lambda_{HS}^{\mathrm{UV}} v_h$, and $g_{saa} = 2\lambda_S v_s$. A physical pole calculation must combine these with the gauge/ghost contributions and a consistent tadpole/VEV prescription. Tadpole and VEV-counterterm terms are local and cancel in the displayed momentum difference, but are required in a complete un-subtracted mass/pole renormalization. Thus this expression fixes the perturbative order and diagram structure; it is not by itself a complete two-loop UV–EFT matched self-energy. We do not add this isolated subset to Π_{hs}^{ret} or use it to classify the near-degenerate region. In the gauge basis

the singlet radial mode has no direct sWW or $s\bar{t}t$ vertex, so direct one-loop W- or top-quark contributions to Π_{hs} are absent at the retained order; portal-induced effects in those loops enter at higher order.

For each charged-lepton flavour, the field-dependent mass matrix and gauge-basis scalar vertices are

$$\mathcal{M}_\ell(h,s) = \begin{pmatrix} 0 & y_\ell(v_h+h)/\sqrt{2} \\ \lambda_\ell(v_s+s)/\sqrt{2} & M_{E_\ell} \end{pmatrix},$$
$$\Gamma_h^\ell = \begin{pmatrix} 0 & y_\ell/\sqrt{2} \\ 0 & 0 \end{pmatrix}, \qquad \Gamma_s^\ell = \begin{pmatrix} 0 & 0 \\ \lambda_\ell/\sqrt{2} & 0 \end{pmatrix}.$$

For the τ interaction the original model contains S^*; its radial real-s vertex is the same for the real benchmark couplings. With

$$U_L^T \mathcal{M}_\ell(0,0) U_R = \mathrm{diag}(m_{\ell 1}, m_{\ell 2}), \qquad G_a^\ell = U_L^T \Gamma_a^\ell U_R \quad (a = h, s),$$

all four internal mass-eigenstate pairs are included for both μ and τ. In the following kernels the flavour label ℓ on G_a^ℓ is suppressed.

Define

$$C_{ab}^{ij} = (G_a)_{ij}(G_b)_{ij}^* + (G_a)_{ji}^*(G_b)_{ji}, \qquad D_{ab}^{ij} = (G_a)_{ij}(G_b)_{ji} + (G_a)_{ji}^*(G_b)_{ij}^*,$$

and

$$K_{ij} = \bar{A}_0(m_i^2) + \bar{A}_0(m_j^2) + (m_i^2 + m_j^2 - p^2)\bar{B}_0(p^2; m_i^2, m_j^2).$$

The one-loop fermion self-energy in the source convention is

$$\Pi_{ab}^{F,\ell}(p^2) = -\frac{1}{16\pi^2} \sum_{i,j=1}^{2} \left[C_{ab}^{ij} K_{ij} + 2 m_i m_j D_{ab}^{ij} \bar{B}_0(p^2; m_i^2, m_j^2) \right].$$

We use $\Gamma_{\mathrm{inv}} = p^2 \mathbf{1} - \mathcal{M}_{\mathrm{loc}}^2 + \Pi_{\mathrm{RC}}$ and $\Pi_B = -\Pi_{\mathrm{RC}}$. The finite functions at $\mu_R = M_E$ are

$$\bar{A}_0(m^2) = m^2 \left(1 - \ln \frac{m^2}{\mu_R^2}\right),$$
$$\bar{B}_0(p^2; m_i^2, m_j^2) = -\int_0^1 dx \, \ln \left[\frac{x m_i^2 + (1-x) m_j^2 - x(1-x)p^2 - i0}{\mu_R^2} \right].$$

The full-theory self-energy contains the local portal-threshold constant and the light-lepton contribution reproduced in the EFT. The momentum-dependent hard term is isolated by

$$\Delta\Pi_{\mathrm{hard}}(m_h^2) = [\Pi_B^{\mathrm{full}}(m_h^2) - \Pi_B^{\mathrm{full}}(0)] - [\Pi_B^{\mathrm{light,exact}}(m_h^2) - \Pi_B^{\mathrm{light,exact}}(0)],$$

where the light-light term uses the exact light-state Yukawa couplings from the same mass diagonalization. The retained EFT soft contribution is the light-lepton bubble plus the dimension-five contact tadpole,

$$\Pi^{B}_{\mathrm{soft,bub+ct}}(p^2) = \sum_{\ell=\mu,\tau} \frac{q_\ell}{16\pi^2} \left[4\bar{A}_0(m_\ell^2) + (4m_\ell^2 - p^2)\bar{B}_0(p^2; m_\ell^2, m_\ell^2)\right],$$
$$q_\ell = \frac{2m_\ell^2}{v_h v_s}.$$

We denote their sum by

$$\Pi^{\mathrm{ret}}_{hs}(m_h^2) = \Pi^{B}_{\mathrm{soft,bub+ct}}(m_h^2) + \Delta\Pi_{\mathrm{hard}}(m_h^2).$$

The soft term here is the specified bubble-plus-contact subset, not the complete set of all same-order EFT soft insertions and counterterms. Thus Π^{ret}_{hs} is a conditional Yukawa-sector benchmark estimate rather than a fully closed model-level pole prediction.

For $M_E = 20\,\mathrm{TeV}$, $\theta_{\mu L} = \theta_{\tau L} = 10^{-2}$, $m_h = 125.25\,\mathrm{GeV}$, $v_h = 246.22\,\mathrm{GeV}$, and $\mu_R = M_E$, the numerical decomposition is

$[\mathrm{GeV}^2]$	Benchmark A	Benchmark B
$\Pi^{B}_{\mathrm{soft,bub+ct}}(m_h^2)$	$-0.437308303 - 0.113173709i$	$-0.098410275 - 0.025468201i$
$\Delta\Pi_{\mathrm{hard}}(m_h^2)$	$+0.024030444$	$+0.005407724$
$\Pi^{\mathrm{ret}}_{hs}(m_h^2)$	$-0.413277859 - 0.113173709i$	$-0.093002551 - 0.025468201i$

The exact hard momentum difference agrees with the leading source-aligned derivative result to 6.11×10^{-4} relative at both benchmarks, and the zero-momentum hard comparator agrees with the leading portal-threshold estimate to 1.79×10^{-4} relative. These checks validate the stated fermionic subcalculation and its numerical implementation.

For a clearly non-degenerate point in Benchmark B, we take $m_s = 100\,\mathrm{GeV}$. Here m_s denotes the unmixed diagonal singlet-mass input in the inverse propagator, not an already diagonalized physical pole mass, and

$$\lambda_S = \frac{m_s^2}{2v_s^2} = 0.05039.$$

The singlet width in the mass interval considered is

$$\Gamma_s = \Theta(m_s - 2m_{A'})\frac{m_s^3}{32\pi v_s^2}(1 - 4x + 12x^2)\sqrt{1-4x}$$
$$+ \sum_{\ell=\mu,\tau} \Theta(m_s - 2m_\ell)\frac{m_\ell^2 m_s}{8\pi v_s^2}\left(1 - \frac{4m_\ell^2}{m_s^2}\right)^{3/2}, \qquad x = \frac{m_{A'}^2}{m_s^2}.$$

At this point $s \to A'A'$ is closed because $m_s < 2m_{A'} = 126\,\mathrm{GeV}$, $h \to A'A'$ is closed because $m_h < 2m_{A'}$, and $h \to ss$ is closed because $m_s > m_h/2$. The width is $\Gamma_s = 0.126811\,\mathrm{MeV}$. The first-pass complex-pole solution gives a singlet-like pole at $99.999999997\,\mathrm{GeV}$ with width $0.126815\,\mathrm{MeV}$, and a Higgs-like pole at $125.250000022\,\mathrm{GeV}$ with width $4.069993\,\mathrm{MeV}$.

For this non-degenerate point, the leading-fermionic retained off-diagonal estimate gives

$$\eta_{hs} \equiv \left| \frac{\Pi^{\mathrm{ret}}_{hs}(m_h^2)}{a-b} \right| \simeq 1.695 \times 10^{-5}, \qquad a = m_h^2 - i m_h \Gamma_h, \quad b = m_s^2 - i m_s \Gamma_s.$$

In the isolated-pole, weak-mixing and SM-dominated-channel limit, the mixing-only leading rate diagnostic is

$$\delta\mu_h^{(\mathrm{mix,LO})} \equiv \mu_h^{(\mathrm{mix,LO})} - 1 \simeq -\eta_{hs}^2 \simeq -2.87 \times 10^{-10}.$$

The corresponding mixing-only fractional rate shift is about $2.9 \times 10^{-8}\%$. Here η_{hs} is a first-order off-diagonal mass-insertion diagnostic, not an independently calculated physical mixing angle. The rate estimate is mixing-only: it is not a complete pole signal strength and does not include process-dependent direct-singlet-Yukawa corrections in the μ and τ channels or the separate searches for the 100 GeV scalar.

For comparison, the PDG 2025 update quotes the inclusive global signal strengths $\mu_{\mathrm{ATLAS}} = 1.023^{+0.056}_{-0.053}$ and $\mu_{\mathrm{CMS}} = 1.014^{+0.055}_{-0.053}$ [17]. Their uncertainties are approximately 5.3%–5.6%, more than 1.8×10^8 times larger than the illustrative 2.9×10^{-10} mixing-only rate shift. This scale comparison concerns the isolated 125 GeV pole in the stated non-degenerate benchmark; the quoted single-narrow-resonance combinations with fixed signal shapes do not provide a model-specific two-scalar likelihood or a bound on the separate S-like state. The present comparison is deliberately restricted to a point with a well-separated S-like pole and with $h \to ss$ closed. For $m_s < m_h/2$, scalar-pair decays such as $h \to ss$, when open, depend on the corresponding trilinear coupling and cannot be inferred from $\cos^2\theta$ alone.

The finite-momentum result does not imply uniformly negligible mixing. The relevant complex inverse propagator is

$$\mathcal{D}_B^{-1}(p^2) = \begin{pmatrix} p^2 - m_h^2 + i m_h \Gamma_h & -\Pi^B_{hs}(p^2) \\ -\Pi^B_{sh}(p^2) & p^2 - m_s^2 + i m_s \Gamma_s \end{pmatrix}, \qquad \det \mathcal{D}_B^{-1}(p^2) = 0.$$

The near-pole solver retains the momentum dependence of the light-lepton soft B_0 functions and analytically continues them to the resonance sheet. Its

hard part uses the source-aligned finite-derivative approximation; at $p^2 = m_h^2$, replacing this by the exact mass-eigenstate hard difference changes the hard term by 6.11×10^{-4} relative and the total retained Π_{hs} by about 3.43×10^{-5} relative. The diagonal real self-energies are absorbed into the benchmark mass inputs and the diagonal widths are held fixed, with $\Gamma_h = 4.07\,\mathrm{MeV}$. The resulting poles are therefore first-pass complex-pole diagnostics, not a complete pole-scheme calculation or an experimental lineshape fit.

At exact diagonal-mass degeneracy $m_s = m_h = 125.25\,\mathrm{GeV}$, the two benchmarks give different pole behavior:

	Benchmark A: $A'A'$ open	Benchmark B: $A'A'$ closed
Γ_s	$3.7468\,\mathrm{GeV}$	$0.15894\,\mathrm{MeV}$
$\lvert\Pi_{hs}/(a-b)\rvert$	9.14×10^{-4}	0.19685
$(m_{\mathrm{pole}}, \Gamma_{\mathrm{pole}})_{\mathrm{narrow}}$	$(125.250001\,\mathrm{GeV}, 4.0727\,\mathrm{MeV})$	$(125.250041\,\mathrm{GeV}, 0.2921\,\mathrm{MeV})$
$(m_{\mathrm{pole}}, \Gamma_{\mathrm{pole}})_{\mathrm{broad}}$	$(125.264006\,\mathrm{GeV}, 3746.39\,\mathrm{MeV})$	$(125.249959\,\mathrm{GeV}, 3.9368\,\mathrm{MeV})$

For the closed-channel benchmark, a $0.1\,\mathrm{MeV}$-step scan over $m_s \in [125.20, 125.30]\,\mathrm{GeV}$ finds a maximum Higgs-like pole-eigenvector composition diagnostic $|s/h| \simeq 0.2039$ near $m_s = 125.2499\,\mathrm{GeV}$; the sampled interval with $|s/h| \geq 0.1$ is approximately 125.2467–$125.2532\,\mathrm{GeV}$. In the open-channel benchmark the maximum over the same local scan is 9.15×10^{-4}, and no point reaches 0.1. The quantity $|s/h|$ is a non-Hermitian pole-eigenvector diagnostic, not a directly observable mixing angle or rate.

Separately, the exact mass-eigenstate fixed-self-energy scan over the local interval $m_s \in [10, 150]\,\mathrm{GeV}$, with $1\,\mathrm{MeV}$ spacing and root-refined threshold crossings, gives no interval satisfying $2|\Pi_{hs}/(a-b)| \geq 0.1$ in Benchmark A. In Benchmark B the corresponding diagnostic interval is

$$m_s \in [125.242553544, 125.257446018]\,\mathrm{GeV},$$

or approximately $|m_s - m_h| \lesssim 7.45\,\mathrm{MeV}$. This is a fixed-self-energy diagnostic, not an allowed or excluded interval. The $|s/h|$ eigenvector criterion and $2|\Pi_{hs}/(a-b)|$ criterion are distinct diagnostics and are not identified with one another. In the narrow near-degenerate region, the single-pole approximation $\mu_h \simeq \cos^2\theta$ is not used; a physical rate requires both pole residues, production and decay amplitudes, interference and the experimental mass response.

The perturbativity condition $\lambda_S = m_s^2/(2v_s^2) \leq 4\pi$ gives

$$m_s \leq m_s^{\max} = \sqrt{8\pi}\, v_s = \begin{cases} 355.4\,\mathrm{GeV}, & \text{Benchmark A}, \\ 1579.2\,\mathrm{GeV}, & \text{Benchmark B}. \end{cases}$$

The numerical interval 10–150 GeV is a local Higgs-pole diagnostic within the stated benchmark, not an exhaustive scan of the full perturbative intervals or a basis for a global scalar-sector exclusion or viability statement. For $m_s \gg m_h$, at fixed benchmark inputs the retained off-diagonal numerator varies slowly with m_s, whereas the diagonal mass separation grows as m_s^2; hence the mixing diagnostic decreases parametrically as $|\Pi_{hs}|/m_s^2$.

For $q^2 \ll m_s^2, m_{A'}^2$, integrating out s and comparing local exchange coefficients gives scalar four-lepton operators with coefficient scale

$$C_{\ell\ell'}^{(s)} \sim \frac{y_{s\ell\ell} y_{s\ell'\ell'}}{m_s^2}, \qquad y_{s\ell\ell} \simeq \frac{m_\ell}{v_s}.$$

Using $m_{A'} = 2 g_X v_s$, the corresponding A'-exchange coefficient scale is $C^{(A')} \sim g_X^2/m_{A'}^2 = 1/(4v_s^2)$, so

$$\frac{|C_{\ell\ell'}^{(s)}|}{|C^{(A')}|} \sim \frac{4 m_\ell m_{\ell'}}{m_s^2}.$$

For $m_s \geq 10\,\text{GeV}$, this coefficient-level ratio is at most 4.5×10^{-4} for a $\mu\mu$ operator; at $m_s = 100\,\text{GeV}$ it is 4.5×10^{-6}. For a $\tau\tau$ operator it can instead be about 0.13 at $m_s = 10\,\text{GeV}$. This scaling comparison therefore supports separating scalar exchange from the muon-dominated effective-current benchmark, but it does not justify neglecting scalar effects in every τ-sensitive process and is not a process-specific rate comparison.

The scalar-suppressed benchmark separates the A' effective-current observables in Fig. 2 from scalar-mediated observables. The radial mode s remains a physical degree of freedom in the UV theory, and its scalar-mediated channels remain part of the model phenomenology; this is a phenomenological separation, not a formal decoupling limit.

Major Points 2).

> *Motivating the range of effective theory parameters. To further aid in the comparison between the UV theory and the two-parameter low-energy theory, it would be useful to explicitly state in one place (perhaps in a table or a figure like Fig. 2) the range of the effective model parameters that are feasible. The main phenomenological results are in Fig. 2, but it was not clear why this range of couplings and A' masses were chosen: are values outside this range not allowed? What, if anything, does the perturbativity of the UV theory impose on the effective parameters?*

Reply. Fig. 2 displays the intermediate-mass laboratory comparison in the window

$$10^{-3}\,\mathrm{GeV} \le m_{A'} \le 10^{2}\,\mathrm{GeV}, \qquad 10^{-6} \le g_X \le 1.$$

The endpoint choices, experimental coverage, and point-dependent UV consistency conditions for this comparison window are explained below.

Laboratory coverage within the displayed window.

Within this window, PIENU probes $m_{A'} < m_{\pi^+} - m_\mu \simeq 33.9\,\mathrm{MeV}$ [18, 19], and the indicative NA62 kaon-decay recast extends to $m_{A'} < m_{K^+} - m_\mu \simeq 388\,\mathrm{MeV}$ [20]. CCFR and CHARM II provide complementary neutrino-trident constraints [21, 22], while NA64$_\mu$ probes the sub-GeV region [23]. BaBar and Belle II cover visible resonances up to approximately 10 and 9 GeV, respectively, and the CMS analysis covers 5–70 GeV [24, 4, 25]. This complementarity motivates the central region displayed.

Lower-mass region.

Below the displayed mass window, astrophysical environments provide important additional sensitivity to weakly coupled light mediators. Supernova cooling and energy transport are particularly relevant to muon- and neutrino-coupled bosons [26, 27], with sensitivity overlapping the sub-GeV part of Fig. 2. Because charged-muon production and absorption differ for an axial current, the full vector $L_\mu - L_\tau$ exclusion is not directly transferable; the axial processes are treated under Major Point 4.4. These astrophysical probes motivate a separate treatment of the lower-mass region alongside the laboratory comparison. Figure 1 presents the separate axial-EFT ballistic-cooling benchmark; under the stated criterion, its calculated contour does not overlap the displayed $g_X \ge 10^{-6}$, $m_{A'} \ge 1$ MeV window. This statement is specific to the stated cooling criterion and approximations; it does not exclude additional diffusive or explosion-energy constraints.

The more direct reason not to extend the figure to much smaller masses is that the $(g-2)_\mu$ bound already dominates the laboratory constraints displayed at its low-mass end, and its low-mass continuation is fixed by a simple asymptotic behavior. In the scalar-suppressed benchmark, for $m_{A'} \ll m_\mu$,

$$\Delta a_\mu^{A'} \simeq -\frac{g_X^2}{4\pi^2}\frac{m_\mu^2}{m_{A'}^2}, \qquad g_X^{\lim} \simeq 1.76\times10^{-6}\left(\frac{m_{A'}}{1\,\mathrm{MeV}}\right),$$

where the second expression uses the adopted approximate 2σ criterion $\Delta a_\mu^{A'} \gtrsim -8.8\times10^{-10}$, obtained from $\Delta a_\mu = 38(63)\times10^{-11}$ [28, 29, 30].

The boundary therefore continues towards smaller couplings in proportion to the mediator mass, appearing as a straight line in the log–log plane. Extending Fig. 2 further to the left would chiefly prolong this already dominant precision-bound regime, rather than reveal an additional transition among the laboratory probes compared here. We therefore keep the figure focused on the interval that displays their complementary reach. As a numerical illustration, the continuation crosses the plotted coupling floor $g_X = 10^{-6}$ at approximately 0.57 MeV. This reasoning assumes the heavy-scalar benchmark, without a cancellation of the negative axial contribution, and does not establish a universal ordering relative to astrophysical bounds.

Upper-mass region.

Above the displayed window, the on-shell meson-decay, B-factory, and Z-decay channels used here have exhausted their kinematic reach. For low-energy virtual exchange, the relevant interaction strength decreases as $g_X^2/m_{A'}^2$ at fixed coupling, and the heavy-mediator contribution to $(g-2)_\mu$ also decouples as $m_\mu^2/m_{A'}^2$. Thus these measurements have progressively weaker sensitivity to higher masses at the couplings of interest, which motivates ending this comparison at 100 GeV. Heavier mediators instead motivate higher-energy probes, including the future muon-collider study discussed under Minor Point 5.

Coupling range and UV perturbativity.

The coupling endpoints 10^{-6} and 1 are display choices: the lower edge retains the low-mass $(g-2)_\mu$ boundary, while the upper edge shows the larger-coupling exclusions. A conventional gauge perturbativity check is $|g_X Q_i| \lesssim \sqrt{4\pi}$ for every charged field, evaluated at the scale in question. For the unit-charge leptons this gives $g_X \lesssim \sqrt{4\pi}$; including the charge-two scalar gives the stronger estimate $g_X \lesssim \sqrt{4\pi}/2 \simeq 1.77$. The plotted upper edge lies within both estimates, but neither criterion selects $g_X = 1$ as an endpoint. Perturbativity of the independent UV Yukawa and scalar couplings must also be checked for a chosen completion; it does not follow from the gauge coupling alone. At leading order, the VLL Yukawa parameters are related to the mixing angles by $\theta_{\ell L} \simeq y_\ell v_h/(\sqrt{2}M_E)$ and $\theta_{\ell R} \simeq \lambda_\ell v_s/(\sqrt{2}M_E)$, as derived under Major Point 1.3. Their perturbativity therefore depends on the chosen M_E, v_s, and mixing-angle inputs; it does not define a universal boundary in the two-parameter $(m_{A'}, g_X)$ plane. The explicit $|\lambda_\ell| \leq \sqrt{4\pi}$ check under Major Point 1.3 applies to its two illustrative portal benchmarks only.

For clarity, the scan variables and their UV interpretation are summarized here:

Quantity	Range or relation	Interpretation
$m_{A'}$	10^{-3}–$10^{2}\,\mathrm{GeV}$	Main laboratory comparison window; lower masses emphasize astrophysical/precision probes, while the searches shown lose reach at higher masses.
g_X	10^{-6}–1	Display range containing the low-mass precision boundary and the larger-coupling exclusions.
v_s	$v_s = m_{A'}/(2g_X)$	Derived from each selected $(m_{A'}, g_X)$ point, not an independent scan variable.
λ_S	$\lambda_S = m_s^2/(2v_s^2) = 2g_X^2 m_s^2/m_{A'}^2$	Scalar self-coupling in the stated small-portal benchmark; its perturbativity condition must be checked separately.
m_s benchmark	$m_s \geq 10\,\mathrm{GeV}$ for the scalar-suppressed reference case	Reference scalar-mass benchmark used to evaluate the UV consistency condition.

Using the one-VEV relation $m_s^2 = 2\lambda_S v_s^2$, the perturbativity condition $\lambda_S \leq \lambda_S^{\max}$ gives

$$m_s \leq \sqrt{2\lambda_S^{\max}}\, v_s.$$

For the illustrative choice $\lambda_S^{\max} = 4\pi$, this becomes $m_s \leq \sqrt{8\pi}\, v_s$. Consequently, the reference benchmark $m_s \geq 10\,\mathrm{GeV}$ requires

$$v_s \geq \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}, \qquad m_{A'} \gtrsim (3.99\,\mathrm{GeV})\, g_X.$$

This is the same $v_s \simeq 2\,\mathrm{GeV}$ boundary shown by the dashed line in Fig. 2. It is a theoretical consistency condition, not an experimental exclusion, for the chosen $m_s \geq 10\,\mathrm{GeV}$, $\lambda_S \leq 4\pi$ benchmark: points with $v_s < 2\,\mathrm{GeV}$ fail this choice, while alternative scalar-sector choices require their own point-dependent λ_S check. Within the displayed range $g_X \leq 1$, the gauge-coupling estimates above do not set the plot boundary.

Interpretation of Fig. 2.

The scalar bounds discussed under Major Point 1 constrain $y_{s\mu\mu}$ directly. They can be translated into the $(m_{A'}, g_X)$ plane only for a specified scalar-mass benchmark and under the UV relation $y_{s\mu\mu} = 2g_X m_\mu / m_{A'}$; they are not direct, benchmark-independent limits on the pure-axial A' current.

For clarity, Fig. 2 should be read according to the following distinctions:

Region or condition	Interpretation
Excluded by a contour shown in Fig. 2	Excluded under the assumptions of that particular effective-current constraint.
Not excluded by the contours shown	Survives the constraints displayed in Fig. 2; this alone does not establish every UV consistency condition.
Fails the chosen scalar benchmark	Does not satisfy $m_s \geq 10\,\mathrm{GeV}$ together with $\lambda_S \leq 4\pi$ at that point; other scalar-sector choices are not ruled out by this condition alone.
Outside the plotted window	Not forbidden by the scan endpoints themselves; no exclusion or viability claim follows solely from being outside the displayed axes.

The table uses the one-VEV matching relation of the original paper. Its two plotting variables parameterize the original A' benchmark; the table is not a scan over additional UV field content or an enlarged completion parameter space.

Major Points 3).

> *Are neutrino oscillations possible? The Standard Model is defined with only left-handed neutrinos because of the possibility that neutrinos are their own antiparticle. In such a case neutrino masses are generated through, e.g. through the Weinberg operator.*
>
> *Thus it is not obvious to me how neutrino oscillations can be included in this framework. If there is a simple solution, the authors should include it. If there is a known less-simple solution in the literature, the authors should cite this. Otherwise, at best this will be a confusing point. At worst, it would make the construction inconsistent with neutrino oscillation experiments.*

Reply. The original one-VEV construction and its possible neutrino-mass completion are distinct. With S as the only scalar that breaks $U(1)_X$, the

original field content leaves a residual Z_2 symmetry and does not produce the observed three-flavor mixing. A renormalizable type-I seesaw completion is obtained by adding a charge-one singlet scalar and three right-handed neutrinos, as in the gauged $L_\mu - L_\tau$ seesaw literature [31, 32, 33]. We discuss the original model and this optional extension separately. The additional fields are not part of the one-VEV benchmark used for the non-neutrino phenomenology.

Reply to Point 3.1: the original one-VEV model

> *In this model, $U_A(1)_{L_{\mu-\tau}}$ invariance allows Weinberg operators for each generation. However, it appears that both the Yukawa matrices and neutrino masses must be aligned and therefore a non-trivial PMNS matrix does not appear to be possible.*

Reply. The original charge assignments, in units $Q_\ell = 1$, are

$$Q_X(L_e, L_\mu, L_\tau) = (0, +1, -1), \qquad Q_X(e_R, \mu_R, \tau_R) = (0, -1, +1),$$
$$Q_X(H) = 0, \qquad Q_X(S) = +2.$$

When only S obtains a VEV, the transformation with gauge parameter $\alpha = \pi$ leaves $\langle S \rangle$ invariant and acts as $(+1, -1, -1)$ on the three lepton families. The residual Z_2 therefore separates the electron family from the (μ, τ) sector.

Define $\chi_\alpha = L_\alpha \cdot H$, with the $SU(2)_L$ and Lorentz contractions understood. The $U(1)_X$ charge of the Weinberg structure $\chi_\alpha \chi_\beta$ is $q_\alpha + q_\beta$, where $q_\alpha = Q_X(L_\alpha)$. The ee and $\mu\tau$ operators are allowed at dimension five. The $\mu\mu$ and $\tau\tau$ entries are allowed with $S^\dagger$ and S insertions, respectively, at dimension six. In contrast, the $e\mu$ and $e\tau$ entries carry odd $U(1)_X$ charge and cannot be generated by any number of S or $S^\dagger$ insertions. The most general neutrino mass texture compatible with the original S-only breaking is therefore

$$M_\nu^{(S)} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & c & d \end{pmatrix}.$$

After diagonalizing the VLL sector, the light charged-lepton mass matrix has the same family block structure,

$$M_\ell^{(S)} = \begin{pmatrix} m_e & 0 & 0 \\ 0 & m_{\mu\mu} & m_{\mu\tau} \\ 0 & m_{\tau\mu} & m_{\tau\tau} \end{pmatrix},$$

where a general μ–τ block is gauge allowed; setting its off-diagonal entries to zero is an additional flavor assumption, not a $U(1)_X$ selection rule. Consequently, the left-handed rotations have the form $U_L = 1 \oplus U_L^{(23)}$ and $U_\nu = 1 \oplus U_\nu^{(23)}$, giving

$$U_{\rm PMNS} = U_L^\dagger U_\nu = {\rm diag}(1, U_{23}),$$

up to unphysical phases and mass ordering. A nontrivial μ–τ rotation is possible, but no electron-flavor admixture is generated. The original model therefore cannot reproduce the observed solar and reactor mixing angles. Moreover, its renormalizable Lagrangian does not specify a neutrino-mass mechanism: $M_\nu^{(S)}$ is the most general effective texture if the allowed Weinberg operators are added, not a prediction of the original UV theory.

Reply to Point 3.2: an optional singlet-scalar type-I seesaw completion

> *Perhaps a full model would require right-handed neutrinos. These would now be charged under $U_A(1)_{L_{\mu-\tau}}$, which likely means the second and third generation must appear as vectorlike combinations. However, it seems to me that mixing between generations is prohibited by $U_A(1)_{L_{\mu-\tau}}$.*

Reply. The residual Z_2 is removed by introducing a complex SM-singlet scalar σ with $Q_X(\sigma) = +1$ and $\langle\sigma\rangle \neq 0$, together with three SM-singlet right-handed neutrinos,

$$Q_X(N_{Re}, N_{R\mu}, N_{R\tau}) = (0, +1, -1).$$

The pair $N_{R\mu}$ and $N_{R\tau}$ carries opposite $U(1)_X$ charges and admits a gauge-invariant Dirac mass $M_{\mu\tau}$; it is vectorlike with respect to the new gauge symmetry. The lepton-doublet and Higgs charges are the same as in the conventional gauged $L_\mu - L_\tau$ construction. The reversed charges of the right-handed charged leptons do not enter the neutrino mass operators.

At the renormalizable level, with the single neutral Higgs doublet, the Dirac Yukawa terms are diagonal:

$$-\mathcal{L}_D = \sum_{\alpha=e,\mu,\tau} y_\alpha^\nu \, \overline{L}_\alpha \widetilde{H} N_{R\alpha} + {\rm h.c.}, \qquad \widetilde{H} = i\sigma_2 H^*,$$

where y^ν_α denotes the neutrino Yukawa coupling. The gauge-invariant Majorana terms are

$$\begin{aligned}-\mathcal{L}_M = &\frac{1}{2}M_{ee}\overline{N^c_{Re}}N_{Re} + M_{\mu\tau}\overline{N^c_{R\mu}}N_{R\tau} \\ &+ h_{e\mu}\sigma^\dagger\overline{N^c_{Re}}N_{R\mu} + h_{e\tau}\sigma\overline{N^c_{Re}}N_{R\tau} \\ &+ \frac{1}{2}h_{\mu\mu}S^\dagger\overline{N^c_{R\mu}}N_{R\mu} + \frac{1}{2}h_{\tau\tau}S\overline{N^c_{R\tau}}N_{R\tau} + \text{h.c.}\end{aligned}$$

The σ VEV generates the $e\mu$ and $e\tau$ entries, while the existing S VEV generates the $\mu\mu$ and $\tau\tau$ entries. The ee and $\mu\tau$ entries are allowed as bare masses. Thus all six independent entries of the complex symmetric Majorana matrix are generated by gauge-invariant terms.

For the VEV convention

$$\langle H^0\rangle = \frac{v_h}{\sqrt{2}}, \qquad \langle S\rangle = \frac{v_s}{\sqrt{2}}, \qquad \langle\sigma\rangle = \frac{v_\sigma}{\sqrt{2}},$$

the Dirac and right-handed Majorana mass matrices are

$$\begin{aligned}m_D &= \frac{v_h}{\sqrt{2}}\,\mathrm{diag}(y^\nu_e, y^\nu_\mu, y^\nu_\tau), \\ M_R &= \begin{pmatrix} M_{ee} & h_{e\mu}v_\sigma/\sqrt{2} & h_{e\tau}v_\sigma/\sqrt{2} \\ h_{e\mu}v_\sigma/\sqrt{2} & h_{\mu\mu}v_s/\sqrt{2} & M_{\mu\tau} \\ h_{e\tau}v_\sigma/\sqrt{2} & M_{\mu\tau} & h_{\tau\tau}v_s/\sqrt{2}\end{pmatrix}.\end{aligned} \tag{1}$$

In the basis (ν_L, N^c_R), the full neutral-fermion mass matrix is

$$\mathcal{M}_{6\times6} = \begin{pmatrix} 0 & m_D \\ m_D^T & M_R\end{pmatrix}. \tag{2}$$

For $\|m_D M_R^{-1}\| \ll 1$, block diagonalization gives the type-I seesaw mass matrix

$$M_\nu \simeq -m_D M_R^{-1} m_D^T. \tag{3}$$

Let U_N denote the Takagi matrix of M_R,

$$U_N^T M_R U_N = \widehat{M}_N, \qquad \widehat{M}_N = \mathrm{diag}(M_1, M_2, M_3).$$

The M_i are the leading heavy-neutrino masses in the seesaw limit, with light–heavy mixing $\Theta \simeq m_D M_R^{-1}$. The light-neutrino mass matrix is diagonalized by

$$U_\nu^T M_\nu U_\nu = \widehat{m}_\nu, \qquad \widehat{m}_\nu = \mathrm{diag}(m_1, m_2, m_3), \tag{4}$$

and the observable mixing matrix is

$$U_{\rm PMNS} = U_L^\dagger U_\nu + \mathcal{O}(\Theta^2),$$

where U_L is the left-handed charged-lepton rotation. Integrating out the heavy N_R fields generates the low-energy Weinberg operator, so the light states are Majorana neutrinos without requiring light right-handed neutrinos.

The construction can accommodate a general nonsingular light-neutrino mass matrix. For a chosen set of light masses and a target PMNS matrix, define

$$U_\nu = U_L U_{\rm PMNS}, \qquad M_\nu^{\rm target} = U_\nu^* \widehat{m}_\nu U_\nu^\dagger.$$

For nonsingular m_D, the corresponding heavy Majorana matrix is

$$M_R = -m_D^T \left(M_\nu^{\rm target}\right)^{-1} m_D. \tag{5}$$

This matrix has precisely the general complex-symmetric form allowed by the terms above. Substitution into the seesaw relation recovers $M_\nu^{\rm target}$. The result is an analytic texture-level construction of the light masses and PMNS mixing; it is not a numerical fit or a global analysis of the extended parameter space.

In a left-handed Weyl basis, the added right-handed neutrinos are represented by N_R^c with charges $(0, -1, +1)$. Their contributions to the linear and cubic $U(1)_X$ anomalies vanish:

$$\Delta\mathcal{A}_{\rm grav^2 X} \propto \sum_i q_i = 0 - 1 + 1 = 0,$$
$$\Delta\mathcal{A}_{X^3} \propto \sum_i q_i^3 = 0^3 + (-1)^3 + (+1)^3 = 0.$$

Since N_R are SM singlets, they introduce no mixed SM-gauge anomalies.

The three Dirac Yukawa terms are invariant separately:

$$Q_X(\overline{L}_e \widetilde{H} N_{Re}) = 0 + 0 + 0 = 0,$$
$$Q_X(\overline{L}_\mu \widetilde{H} N_{R\mu}) = -1 + 0 + 1 = 0,$$
$$Q_X(\overline{L}_\tau \widetilde{H} N_{R\tau}) = +1 + 0 - 1 = 0.$$

The scalar-assisted Majorana terms are also neutral,

$$Q_X(\sigma^\dagger \overline{N_{Re}^c} N_{R\mu}) = -1 + 0 + 1 = 0,$$
$$Q_X(\sigma \overline{N_{Re}^c} N_{R\tau}) = +1 + 0 - 1 = 0,$$
$$Q_X(S^\dagger \overline{N_{R\mu}^c} N_{R\mu}) = -2 + 1 + 1 = 0,$$
$$Q_X(S \overline{N_{R\tau}^c} N_{R\tau}) = +2 - 1 - 1 = 0.$$

The bare terms satisfy

$$Q_X(\overline{N_{Re}^c}N_{Re}) = 0, \qquad Q_X(\overline{N_{R\mu}^c}N_{R\tau}) = +1 - 1 = 0.$$

Thus all entries arise from gauge-invariant terms, and the off-diagonal entries are generated by spontaneous $U(1)_X$ breaking.

The field assignments and neutrino interactions correspond to the two-singlet-scalar construction of Ref. [32]: see Table I and Eq. (4), together with the displayed M_R and M_D matrices and the seesaw relation in Eq. (8). The same structure is given in Sec. II, Eqs. (1)–(5), of Ref. [33], after identifying their charge-one and charge-two singlets (Φ_1, Φ_2) with (σ, S) and matching the VEV conventions.

The single-charge-one-scalar construction of Ref. [34] has $(M_R)_{\mu\mu} = (M_R)_{\tau\tau} = 0$. For diagonal nonsingular m_D, the seesaw relation then gives

$$M_\nu^{-1} = -m_D^{-T} M_R m_D^{-1}, \qquad (M_\nu^{-1})_{\mu\mu} = (M_\nu^{-1})_{\tau\tau} = 0.$$

This two-zero-minor condition is not imposed in the present construction: the existing charge-two scalar S generates both diagonal entries of the μ–τ block of M_R.

With canonical gauge kinetic terms, the two singlet VEVs give

$$m_{A'}^2 = g_X^2(4v_s^2 + v_\sigma^2). \tag{6}$$

Defining

$$r_\sigma \equiv \frac{v_\sigma^2}{4v_s^2},$$

the symmetry-breaking scale is

$$v_s = \frac{m_{A'}}{2g_X\sqrt{1 + r_\sigma}}.$$

The original one-VEV relation $m_{A'} = 2g_X v_s$ is recovered for $r_\sigma = 0$. This limit also sets $v_\sigma = 0$ and restores the residual Z_2, so it does not provide the three-flavor mixing obtained in the completion.

The renormalizable scalar potential contains, in addition to the original H- and S-dependent terms,

$$\begin{aligned}\Delta V = & -\mu_\sigma^2|\sigma|^2 + \lambda_\sigma|\sigma|^4 + \lambda_{H\sigma}|H|^2|\sigma|^2 \\ & + \lambda_{S\sigma}|S|^2|\sigma|^2 + \left(\kappa S^\dagger \sigma^2 + \text{h.c.}\right),\end{aligned}$$

where κ has mass dimension one. The last term is gauge invariant and fixes the relative phase of the two singlet VEVs, removing the otherwise accidental relative phase symmetry. The S and σ radial modes can mix; hence $m_s^2 = 2\lambda_S v_s^2$ is not used as a general scalar-mass relation in the extended theory.

The additional scalar also permits the charged-lepton–VLL interactions $\sigma \overline{E}_{\mu L} e_R$ and $\sigma^\dagger \overline{E}_{\tau L} e_R$. Their coefficients are suppressed when the flavor-diagonal charged-lepton benchmark is imposed. The $\sigma + 3N_R$ fields therefore provide a gauge-invariant neutrino-mass completion, while the non-neutrino phenomenology in the manuscript continues to refer to the original one-VEV $S +$ VLL model. **4)**.

> *Is this the full set of experimental constraints? The second half of the manuscript presents a set of experimental bounds on the two-parameter effective A' model. It is not obvious if this is the full set of constraints on the model. (See also point #1 above: I would have liked to see a section discussing the bounds on the S within this model since it cannot be decoupled.) The following experimental bounds seem like they could be relevant. It would be helpful to either include them or justify why they are subdominant to the experiments presented.*

Reply.

The contours in Fig. 2 constrain the specified two-parameter pure-axial effective-current benchmark; they are not presented as an exhaustive global classification of every point in the UV model. We address the additional issues raised here by summarizing the scalar constraints under Major Point 1, estimating the momentum-dependent kinetic mixing, quantifying the leading effects of the finite vector component, and distinguishing the supernova contributions governed by the unchanged neutrino current from those sensitive to the charged-muon current. The conclusions below are stated for the explicit benchmark and mass regimes in which the corresponding comparison applies.

4.1).

> *Neutrino oscillation observations. See point #3 above. It appears that this model predicts a unit PMNS matrix, which is inconsistent with neutrino oscillations.*

Reply.

The original S-only model does not predict a unit PMNS matrix in general; it predicts the restricted form $U_{\rm PMNS} = \mathrm{diag}(1, U_{23})$, as derived under Major Point 3.1. This allows μ–τ mixing but no electron-flavor admixture, and therefore cannot reproduce the observed solar and reactor mixing angles.

The optional $\sigma{+}3N_R$ completion under Major Point 3.2 breaks the residual Z_2 and generates a general light-neutrino mass matrix through the type-I seesaw. Its mass matrices, seesaw relation, Takagi diagonalization, and PMNS matrix are given in Eqs. (1)–(5). This completion is a separate neutrino-sector extension; it does not alter the original one-VEV benchmark used for the non-neutrino phenomenology. In particular, the two-VEV gauge-boson mass relation in Eq. (6) applies only to the optional completion.**4.2)**.

> *There is no quantitative estimate of the amount of kinetic mixing generated with the "benchmark" choices of $\theta \leq 10^{-2}$. The authors make the insightful point that kinetic mixing goes to zero in the axial limit, but because this limit is tied to the size of the mixing (the mass of the vectorlike leptons), for the above benchmark there is non-zero kinetic mixing. It would be useful to have parametric estimates of the amount of kinetic mixing one expects so that readers may compare to experimental bounds. One could invoke parametric expressions in the literature for the relevant loop diagrams.*

Reply.

Finite light–heavy lepton mixing generates a nonzero photon–A' mixing of order $g_X\theta_R^2$. With equal right-handed mixing angles, degenerate physical heavy-lepton masses, and zero independent matching coefficient, $\theta_R = 10^{-2}$ gives $|\epsilon_{\gamma A'}(0)| = 1.44 \times 10^{-6}|g_X|$. At $m_{A'} = 100\,\mathrm{MeV}$ and $g_X = 0.1$, its on-shell value is 1.49×10^{-7}, well below the $\mathcal{O}(10^{-4})$ sensitivity of the NA64 electron-beam invisible search [35]. The derivation and the conditions for this experimental comparison are given below.

One-loop origin and matching prescription.

We define $\epsilon_{\gamma A'}$ in the photon convention, with induced interaction $\mathcal{L} \supset e\epsilon_{\gamma A'} A'_\alpha J^\alpha_{\rm EM}$ after diagonalizing the kinetic terms. It is this coefficient, rather than a hypercharge-mixing coefficient before electroweak symmetry breaking,

that enters the dark-photon comparison. In the muon block the chiral couplings are $g_L^\mu = g_X$ and $g_R^\mu = -g_X \cos 2\theta_{\mu R}$. Consequently,

$$g_V^\mu = \frac{g_L^\mu + g_R^\mu}{2} = g_X \sin^2\theta_{\mu R}, \qquad g_A^\mu = \frac{g_R^\mu - g_L^\mu}{2} = -g_X \cos^2\theta_{\mu R}.$$

The tau block has the opposite charge signs. At one loop, the vector–axial two-point function vanishes in vacuum: its parity-odd tensor is proportional to $\varepsilon^{\alpha\beta\rho\sigma} q_\rho q_\sigma = 0$. Thus the parity-even photon kinetic term depends on g_V, not on the unsuppressed axial coupling. The photon vertex remains diagonal and proportional to the identity in each light–heavy block, so a single off-diagonal A' vertex cannot close this two-point fermion loop.

The heavy threshold can be exhibited in the same mass basis. Writing $s_\ell = \sin\theta_{\ell R}$, $c_\ell = \cos\theta_{\ell R}$, the four diagonal vector couplings are

$$(g_V^\mu, g_V^\tau, g_V^{E_\mu}, g_V^{E_\tau}) = g_X(s_\mu^2, -s_\tau^2, c_\mu^2, -c_\tau^2).$$

For these leptons of electric charge -1, the one-loop photon mixing in the above convention can be written as

$$\epsilon_{\gamma A'}(q^2) = \epsilon_{\rm ind} + \frac{e}{2\pi^2} \sum_{f=\mu,\tau,E_\mu,E_\tau} g_V^f \int_0^1 dx\, x(1-x) \ln \frac{m_f^2 - x(1-x)q^2 - i0}{\mu_{\rm r}^2}.$$

Here m_f are physical mass eigenvalues, $\mu_{\rm r}$ is a renormalization scale, and $\epsilon_{\rm ind}$ is the independent local mixing coefficient. Since $\sum_f g_V^f = 0$, the common ultraviolet term and the explicit $\mu_{\rm r}$ dependence cancel in this sum; that cancellation does not fix $\epsilon_{\rm ind}$. We impose the matching condition

$$\boxed{\epsilon_{\rm ind}(\mu_{\rm match}) = 0}$$

at the heavy-VLL matching scale. This boundary condition applies only to the independent local coefficient; it does not set the full momentum-dependent physical mixing $\epsilon_{\gamma A'}(q^2)$ to zero, because finite light-lepton loop contributions remain. For equal right-handed angles the heavy contribution at zero momentum is

$$\epsilon_{\rm heavy}(0) = \frac{e g_X \cos^2\theta_R}{12\pi^2} \ln \frac{m_{E_\mu}^2}{m_{E_\tau}^2}.$$

It vanishes for degenerate physical heavy states; their full loop functions then cancel at any momentum. Equal coupling weights are also required, and equality of the bare mass parameters alone need not ensure exact degeneracy

after lepton mixing. We take this paired-heavy-state reference choice together with the matching condition above. These assumptions leave a finite light-lepton contribution because $m_\mu \neq m_\tau$.

The difference from imposing only the original common bare mass M_E can also be estimated. In the small-mixing expansion,

$$m_{E_\ell}^2 \simeq M_E^2(1+\theta_{\ell L}^2+\theta_{\ell R}^2), \qquad \epsilon_{\text{heavy}}(0) \simeq \frac{e g_X}{12\pi^2}(\theta_{\mu L}^2-\theta_{\tau L}^2),$$

where the second expression keeps equal right-handed angles and $\epsilon_{\text{ind}} = 0$. With $|\theta_{\ell L}| \leq 10^{-2}$, this residual threshold obeys $|\epsilon_{\text{heavy}}(0)| \lesssim 2.56 \times 10^{-7}|g_X|$, to leading order in the mixing angles. Its momentum dependence is negligible at 100 MeV for $M_E > 10$ TeV. Thus the physical-mass-degenerate reference and the common-bare-mass choice can be compared without assuming that their thresholds are identical.

Finite light-lepton loop and momentum dependence.

For the light-lepton loop, we use the momentum-transfer-dependent kinetic-mixing function studied by Hapitas, Tuckler, and Zhang [36], replacing their charged-lepton vector coupling by $g_X \sin^2\theta_R$ and analytically continuing their spacelike expression with $Q^2 \to -q^2 - i0$. The timelike threshold behavior is compared with Wang and Zhang [27]. Under the matching assumptions above, the remaining contribution is

$$\epsilon_{\gamma A'}(q^2) = -\frac{e\, g_X \sin^2\theta_R}{2\pi^2}\int_0^1 dx\, x(1-x)\log\left[\frac{m_\tau^2 - x(1-x)q^2 - i0}{m_\mu^2 - x(1-x)q^2 - i0}\right],$$

where $\theta_{\mu R} = \theta_{\tau R} \equiv \theta_R$. The light-lepton ultraviolet terms cancel separately in this limit. At $q^2 = 0$, $\int_0^1 dx\, x(1-x) = 1/6$, giving

$$\epsilon_{\gamma A'}(0) = -\frac{e\, g_X \sin^2\theta_R}{12\pi^2}\ln\frac{m_\tau^2}{m_\mu^2} = -\frac{e\, g_X \sin^2\theta_R}{6\pi^2}\ln\frac{m_\tau}{m_\mu}.$$

Using $e = 0.3028$, $m_\mu = 105.658$ MeV, $m_\tau = 1.77686$ GeV, and $\theta_R = 10^{-2}$, the zero-momentum magnitude is

$$\begin{aligned}|\epsilon_{\gamma A'}(0)| &\simeq 1.44 \times 10^{-6}|g_X| \quad (\theta_R = 10^{-2}),\\ &\simeq 1.44 \times 10^{-7}\left(\frac{|g_X|}{0.1}\right)\left(\frac{\theta_R}{10^{-2}}\right)^2 \quad \text{for small } \theta_R.\end{aligned}$$

At fixed off-diagonal lepton mass entry m_R, $\theta_R \simeq m_R/M_E$, so this contribution scales as $g_X m_R^2/M_E^2$. This explicitly exhibits its decoupling with the vector-like-lepton mass under that fixed-parameter choice; maintaining the observed light-lepton masses can require correlated changes of the Yukawa parameters.

For an on-shell A', the relevant timelike argument is $q^2 = m_{A'}^2$. The loop function becomes complex above the charged-lepton pair thresholds; its magnitude is not constant across the scan. A numerical evaluation of the integral gives the following representative values for $\theta_R = 10^{-2}$:

$m_{A'}$	$\lvert\epsilon_{\gamma A'}(m_{A'}^2)\rvert$ / $\lvert g_X\rvert$	$\lvert\epsilon_{\gamma A'}(m_{A'}^2)\rvert$ / $\lvert\epsilon_{\gamma A'}(0)\rvert$
10 MeV	1.44×10^{-6}	1.00
100 MeV	1.49×10^{-6}	1.04
200 MeV	1.82×10^{-6}	1.26
212 MeV	2.12×10^{-6}	1.47
300 MeV	1.74×10^{-6}	1.21
500 MeV	1.40×10^{-6}	0.97
1 GeV	1.08×10^{-6}	0.75
3 GeV	8.06×10^{-7}	0.56
10 GeV	5.47×10^{-8}	3.79×10^{-2}
100 GeV	4.84×10^{-10}	3.35×10^{-4}

The value near $m_{A'} = 2m_\mu$ agrees with the approximately 1.47 enhancement of the magnitude over its zero-momentum value reported by Wang and Zhang [27]. For the low-mass benchmark, the induced electron-current coupling is of order $e|\epsilon| \simeq 4.4 \times 10^{-7}|g_X|$. The momentum-dependent expression should be used for on-shell production and decay; spacelike virtual exchange requires evaluating the same function at the relevant spacelike q^2.

For the reference normalization $g_X = 0.1$, the zero-momentum value is 1.44×10^{-7}, and the largest tabulated on-shell value is 2.12×10^{-7}. The displayed range in Fig. 2 extends to $g_X = 1$, so the corresponding conservative upper estimates over that rectangle are ten times larger.

NA64 comparison and decay branching fractions.

The appropriate dark-photon comparison is the NA64 electron-beam missing-energy search [35], rather than its thermal-dark-matter parameter combinations. The experiment constrains the electromagnetic production vertex $e\epsilon_{\gamma A'} A'_\alpha J^\alpha_{\mathrm{EM}}$, which is precisely the vertex induced by the photon–A' kinetic mixing here. Its Fig. 3 gives a 90% confidence-level exclusion boundary in the $(m_{A'}, \epsilon)$ plane from 9.37×10^{11} electrons on target. We compare to that mass-dependent boundary, not to the thermal-relic parameters y, α_D, or m_χ.

In the original model, neglecting neutrino masses, the leading widths

below the dimuon threshold are

$$\Gamma_{\nu\bar{\nu}} = \frac{g_X^2 m_{A'}}{12\pi}, \qquad \Gamma_{e^+e^-} = \frac{e^2|\epsilon_{\gamma A'}(m_{A'}^2)|^2 m_{A'}}{12\pi}(1+2r_e)\sqrt{1-4r_e},$$

where the first expression sums the two charged neutrino flavors and $r_e = m_e^2/m_{A'}^2$; the electron width is zero below threshold. In the original model, this sum runs over the two massless active-flavor channels ν_μ and ν_τ, with $U(1)_X$ charges $+1$ and -1, respectively; the electron flavor is uncoupled. At $m_{A'} = 100\,\mathrm{MeV}$ and $\theta_R = 10^{-2}$, $\Gamma_{e^+e^-}/\Gamma_{\nu\bar{\nu}} \simeq 2.05 \times 10^{-13}$, so $\mathrm{Br}_{\mathrm{inv}} \simeq 1$.

Above the dimuon threshold, the direct axial muon current opens a visible channel. In the same convention,

$$\Gamma_{\mu^+\mu^-} = \frac{m_{A'}}{12\pi}\sqrt{1-4r_\mu}\left[(g_V^\mu)^2(1+2r_\mu) + (g_A^\mu)^2(1-4r_\mu)\right], \qquad r_\mu = \frac{m_\mu^2}{m_{A'}^2}.$$

At $m_{A'} = 300\,\mathrm{MeV}$, with $g_V^\mu = g_X \sin^2\theta_R$, $g_A^\mu = -g_X\cos^2\theta_R$, and $\theta_R = 10^{-2}$, this gives

$$\frac{\Gamma_{\mu^+\mu^-}}{\Gamma_{\nu\bar{\nu}}} \simeq 0.3576, \qquad \mathrm{Br}_{\mathrm{inv}} \simeq 0.7366.$$

The loop-induced electromagnetic widths, including the hadronic width above the two-pion threshold, are negligible relative to these direct widths at the stated benchmark. We therefore compare the missing-energy signal using

$$\epsilon_{\mathrm{eff}}(m_{A'}) = |\epsilon_{\gamma A'}(m_{A'}^2)|\sqrt{\mathrm{Br}_{\mathrm{inv}}(m_{A'})}, \qquad N_{\mathrm{sig}} \propto \epsilon_{\mathrm{eff}}^2,$$

assuming the published missing-energy acceptance for an invisible spin-one state. This is a leading-yield rescaling, not a detector-level recast of visible dimuon decays.

For a transparent mass-dependent comparison, we read the NA64 90% CL boundary from Fig. 3 of Ref. [35] in log–log coordinates. The figure does not provide a machine-readable table; we therefore quote conservative visual-read-off intervals, with an approximately 0.2–0.3 dex line-thickness/read-off uncertainty, rather than spurious high-precision point values. The lower edge of each interval is used in the ratios below. The model values are evaluated from the full timelike loop function at the same mass:

$m_{A'}$	$\lvert\epsilon\rvert/\lvert g_X\rvert$	$\mathrm{Br}_{\mathrm{inv}}$	$\epsilon^{90\%}_{\mathrm{NA64}}$ (visual range)	ϵ_{eff} at $g_X = 0.1$	R at $g_X = 0.1$
10 MeV	1.44×10^{-6}	1.000	$(0.8\text{–}2.0) \times 10^{-5}$	1.44×10^{-7}	1.80×10^{-2}
100 MeV	1.49×10^{-6}	1.000	$(0.5\text{–}1.2) \times 10^{-4}$	1.49×10^{-7}	2.99×10^{-3}
200 MeV	1.82×10^{-6}	1.000	$(0.7\text{–}1.8) \times 10^{-4}$	1.82×10^{-7}	2.59×10^{-3}
300 MeV	1.74×10^{-6}	0.737	$(1.0\text{–}2.5) \times 10^{-4}$	1.50×10^{-7}	1.50×10^{-3}

Here $R \equiv \epsilon_{\rm eff}/\epsilon_{\rm NA64,low}$, where $\epsilon_{\rm NA64,low}$ is the conservative lower edge of the visually read-off interval. The corresponding signal-yield ratios R^2 are 3.3×10^{-4}, 8.9×10^{-6}, 6.7×10^{-6}, and 2.2×10^{-6}, respectively, at $g_X = 0.1$. Since $\epsilon \propto g_X$, the amplitude ratios are ten times larger and the yield ratios one hundred times larger at $g_X = 1$; even there the largest conservative ratio is $R \simeq 0.18$ at 10 MeV, corresponding to about 3.3% of the limiting signal yield. The other three mass points remain below 10^{-3} of the limiting yield at $g_X = 1$. These upper-edge values are a conservative comparison over the plotted rectangle, not a claim that those large-coupling points satisfy every other constraint.

The secondary-positron resonance energy is $E_{e^+}^{\rm res} = m_{A'}^2/(2m_e)$: it is approximately 0.098, 9.78, and 39.1 GeV at 10, 100, and 200 MeV, respectively, below the approximately 50 GeV missing-energy selection; at 300 MeV it is approximately 88.1 GeV, so the published NA64 curve includes the resonant secondary-positron contribution as well as bremsstrahlung. The rescaling by $\mathrm{Br}_{\rm inv}$ accounts for the open dimuon width at this point. For $g_X = 0.1$ and 1 the total width remains narrow compared with the mass. No dark-matter particle or thermal-relic assumption is required; the invisible final states here are neutrinos.

The zero-momentum light-lepton loop gives $|\epsilon_{\gamma A'}(0)| = 1.44 \times 10^{-6}|g_X|$, whereas the paired-heavy-mass on-shell light-lepton contribution at $m_{A'} = 100\,\mathrm{MeV}$ is $1.49 \times 10^{-6}|g_X|$. The residual common-bare-mass threshold contribution is at most $2.56 \times 10^{-7}|g_X|$. Applying the triangle inequality, and using the negligible momentum dependence of this heavy-threshold term at 100 MeV for $M_E > 10\,\mathrm{TeV}$, gives the conservative leading-order on-shell bound

$$\left|\epsilon_{\gamma A'}(m_{A'}^2)\right|_{m_{A'}=100\,\mathrm{MeV}} \leq (1.49 + 0.256) \times 10^{-6}|g_X| \simeq 1.75 \times 10^{-6}|g_X|.$$

The mass-dependent table uses the paired physical-heavy-mass reference; this residual-threshold estimate does not change the conclusion that the induced electron-production channel is below the published NA64 sensitivity at the representative mass points. This comparison is separate from the direct muon-production constraint of $\mathrm{NA64}_\mu$ in Fig. 2.

Unequal right-handed mixing angles, independent heavy-mass splittings, or a nonzero independent matching coefficient require evaluating the general four-state expression above.

4.3).

Another way of reframing the above point: the bounds presented

are in the pure axial limit. Since the authors present a UV model that only realizes the axial coupling in a decoupling limit, it would be useful to comment on possible experimental bounds on the small vector-like coupling in a model with non-infinite vectorlike fermion masses. If these bounds are always small compared to the axial bounds, this should be stated and justified. The importance of this point is as follows: often bounds are sensitive to the chiral structure of interactions. Sometimes a vector interaction is more tightly constrained than its axial counterpart. Since the authors focus on the phenomenology in the axial limit, one wants to make sure that any small-but-finite vector interaction isn't significant because the constraints on the coupling are orders-of-magnitude larger than the axial coupling.

Reply.

Within the adopted finite-mixing domain, the vector contribution is subleading in neutrino trident production and $(g-2)_\mu$, while the high-mass four-muon comparison and the dimuon-threshold behavior are quantified below. These results follow from the common-g_X coupling hierarchy and the process-specific comparisons.

Finite-mixing domain and vector-to-axial hierarchy.

We restrict the UV parameter region underlying the axial-vector phenomenology to the finite-mixing domain

$$|\theta_{\ell L}|,\ |\theta_{\ell R}| \leq 10^{-2}, \qquad \ell = \mu, \tau,$$

under the small-mixing condition adopted for the UV construction. This is a finite-mass domain, not the strict $M_E \to \infty$ limit: in the seesaw-like regime,

$$m_\ell \simeq \theta_{\ell L}\theta_{\ell R} M_{E\ell}.$$

For the degenerate-VLL reference choice, saturating both angle upper values in the τ sector corresponds to $M_E \simeq 17.8$ TeV; smaller mixing angles correspond to larger, still finite VLL masses. In the mass basis, the diagonal light-lepton couplings retain the opposite overall $U(1)_X$ signs of the two flavor sectors:

$$\begin{aligned}(g_V^\mu, g_A^\mu) &= g_X\big(\sin^2\theta_{\mu R}, -\cos^2\theta_{\mu R}\big),\\ (g_V^\tau, g_A^\tau) &= -g_X\big(\sin^2\theta_{\tau R}, -\cos^2\theta_{\tau R}\big).\end{aligned}$$

Their absolute vector-to-axial ratios are nevertheless the same functions of the corresponding right-handed angles,

$$\left|\frac{g_V^\ell}{g_A^\ell}\right| = \tan^2\theta_{\ell R} \leq 1.01\times 10^{-4}, \qquad \ell = \mu, \tau.$$

Thus the magnitudes of the vector couplings are suppressed by at least four orders of magnitude relative to the corresponding axial couplings throughout the parameter region considered. At the reference value $g_X = 0.1$, this gives $|g_V^\ell| \leq 10^{-5}$; at the plotting upper edge $g_X = 1$, it gives $|g_V^\ell| \leq 10^{-4}$. The neutrino couplings remain $g_{\nu_\mu} = +g_X$ and $g_{\nu_\tau} = -g_X$.

Criterion for vector-specific experimental bounds.

The angle restriction also gives a direct criterion for assessing a vector-specific experimental bound. At a fixed $m_{A'}$, let $g_{V,\mathrm{lim}}$ and $g_{A,\mathrm{lim}}$ denote upper limits on the magnitudes of the corresponding vector and axial charged-lepton couplings, obtained with matched production, decay, and acceptance assumptions. Since both couplings are proportional to the same g_X, the vector limit is weaker than the axial limit when

$$\frac{g_{V,\mathrm{lim}}}{g_{A,\mathrm{lim}}} > \tan^2\theta_{\ell R}.$$

Equivalently, the angle required for the vector-specific bound to remain subdominant is

$$|\theta_{\ell R}| < \arctan\sqrt{\frac{g_{V,\mathrm{lim}}}{g_{A,\mathrm{lim}}}}.$$

For our adopted upper value $|\theta_{\ell R}| = 10^{-2}$, a vector-specific limit would have to constrain the vector coupling to below approximately 10^{-4} of the corresponding axial limit—more than four orders of magnitude tighter at the same mass—to yield a stronger bound on g_X. This is the quantitative criterion for the small-mixing region used here.

For neutrino trident production, the ratio of the leading Standard-Model interference terms induced by the vector and axial components is parametrically

$$\left|\frac{C_V\delta C_V}{C_A\delta C_A}\right| \sim \frac{C_V}{C_A}\tan^2\theta_{\mu R} \lesssim 2.0\times 10^{-4}, \qquad C_V \simeq 0.96, \quad C_A = \frac{1}{2},$$

at the same momentum transfer, before the distinct finite-m_μ response functions are included. For the muon anomalous magnetic moment,

$$\left|\frac{\Delta a_\mu^V}{\Delta a_\mu^A}\right| = \tan^4\theta_{\mu R}\frac{F_V(r)}{F_A(r)} \lesssim 2.1\times 10^{-9},$$

using $F_V/F_A \leq 0.20$ over the mass range considered. These are ratios of contributions to the specified observables, not standalone experimental exclusion limits; they show that the finite vector component is subleading in the trident-interference and $(g-2)_\mu$ observables used in our analysis.

Comparison with the four-muon searches.

For the BaBar, Belle II, and CMS four-muon final-state-radiation searches used in Fig. 2, the unit-coupling vector and axial production kernels become nearly identical for $m_{A'} \gg m_\mu$, with correspondingly similar acceptances; away from threshold, the dimuon widths and branching fractions also approach the same leading behavior [24, 4, 25]. Thus the vector-only coupling reach in these same channels is of the same order as the axial reach, rather than more than four orders of magnitude stronger. In the mixed current, the separate vector-current-squared production contribution is weighted by

$$\left|\frac{g_V^\mu}{g_A^\mu}\right|^2 = \tan^4\theta_{\mu R} \leq 1.01 \times 10^{-8},$$

in this away-from-threshold regime. The dimuon branching fraction still requires the threshold-dependent width factors given below.

The loop-induced electromagnetic coupling associated with the same vector admixture is treated separately through the kinetic-mixing estimate in our response to Major Point 4.2.

Threshold-dependent dimuon widths.

For visible dimuon decays, the vector and axial partial widths have different threshold behavior:

$$\Gamma_V \propto (g_V^\mu)^2(1+2r_\mu)\sqrt{1-4r_\mu}, \qquad \Gamma_A \propto (g_A^\mu)^2(1-4r_\mu)^{3/2}, \quad r_\mu = \frac{m_\mu^2}{m_{A'}^2}.$$

Their ratio is therefore

$$\frac{\Gamma_V}{\Gamma_A} = \tan^4\theta_{\mu R}\,\frac{1+2r_\mu}{1-4r_\mu}.$$

For any chosen tolerance η on the relative vector contribution, the corresponding angle condition is mass dependent:

$$|\theta_{\mu R}| < \arctan\left[\eta^{1/4}\left(\frac{1-4r_\mu}{1+2r_\mu}\right)^{1/4}\right].$$

Away from threshold, the vector-squared contribution is suppressed by $\tan^4\theta_{\mu R} \leq 1.01 \times 10^{-8}$. Arbitrarily close to $m_{A'} = 2m_\mu$, however, the axial phase-space factor vanishes more rapidly, so no nonzero mass-independent upper bound on the mixing angle can keep the vector-to-axial width ratio uniformly small arbitrarily close to threshold.

Thus, under the adopted small-mixing condition $|\theta_{\ell R}| \leq 10^{-2}$ and away from the dimuon threshold, the process-specific estimates show that the finite vector component is subleading in neutrino trident production and $(g-2)_\mu$, while the vector-only and axial-only coupling reaches in the high-mass four-muon searches are of the same order. Near $m_{A'} = 2m_\mu$, the visible dimuon rate uses the finite-angle widths above.

4.4).

> *Are stellar cooling bounds not relevant? See e.g. 2511.09619 for a recent paper on this. If they are not, then this is certainly worth justifying.*

Reply.

Supernova cooling is relevant to the sub-GeV region. The referee-cited analysis is now published as JHEP 06 (2026) 155 [26]; a recent update for the conventional vector $L_\mu - L_\tau$ boson, using the SFHo-18.8 profile, reports SN1987A sensitivity up to $m_{A'} \sim 300$ MeV and low-energy-supernova sensitivity up to about 600 MeV, with SN1987A coupling sensitivity of order 10^{-10} and low-energy-supernova sensitivity reaching 6.4×10^{-12} [27]. These values provide reference scales for assessing the axial benchmark. Kahn et al. give an earlier discussion of stellar cooling for ultralight axial forces with $m_{A'} < 2m_e$: their Sec. 2.4 emphasizes that an axial A' can have unsuppressed neutrino couplings and a stellar production mechanism not necessarily dominated by kinetic mixing, and that a dedicated analysis is required [37]. This motivates treating the axial case separately.

Neutrino-coalescence production and invisible decays.

In our model, the tree-level neutrino couplings are the same as in the conventional $L_\mu - L_\tau$ model,

$$g_{\nu_\mu} = +g_X, \qquad g_{\nu_\tau} = -g_X.$$

The flavor-summed neutrino-coalescence matrix element and the invisible partial widths $A' \to \nu_\alpha \bar{\nu}_\alpha$ therefore have the same coupling normalization as

in the conventional vector $L_\mu - L_\tau$ reference at fixed $(m_{A'}, g_X)$, for the same neutrino phase-space distributions and supernova profile. This follows directly from the original model's active-flavor gauge current $g_{\nu_\mu} = +g_X$, $g_{\nu_\tau} = -g_X$; no additional neutrino fields or completion assumptions are used in this current-level comparison. In the cited supernova profile, neutrino coalescence dominates the production luminosity for $m_{A'} \gtrsim 10$ MeV. Accordingly, for 10 MeV $\lesssim m_{A'} < 2m_\mu$, the published SN1987A cooling result is a useful reference scale: neutrino-coalescence production is unchanged and dominates the luminosity in the cited profile, while A' decays predominantly to neutrinos. We use this as a benchmark comparison, not as a fully recast exclusion contour for the axial model. Equality of the neutrino-coalescence matrix element and invisible partial-width normalization is a current-level statement; it does not imply equality of the complete supernova cooling or energy-transport limits. Those limits also depend on inverse absorption, optical depths and trapping, decay and energy deposition, and the stellar profile; the charged-muon axial and longitudinal channels discussed below can change these parts of the transport problem.

Charged-muon axial semi-Compton production and absorption.

The distinction arises in processes involving the charged-muon current. The semi-Compton process $\mu\gamma \to \mu A'$ and its inverse absorption depend on whether that current is vector or axial. For on-shell muons, the vector current is conserved whereas the axial current obeys

$$q_\alpha \bar{u}(p')\gamma^\alpha u(p) = 0, \qquad q_\alpha \bar{u}(p')\gamma^\alpha \gamma_5 u(p) = 2m_\mu \bar{u}(p')\gamma_5 u(p), \qquad q = p' - p.$$

The non-conserved axial current therefore contributes to longitudinal A' emission. In the relativistic longitudinal limit $E_{A'} \gg m_{A'}$, Goldstone equivalence gives the effective coupling scale

$$|g_{G\mu}| \sim \frac{2|g_A^\mu| m_\mu}{m_{A'}}.$$

The semi-Compton emissivity and inverse absorption must consequently be evaluated with the axial current and its polarization structure; the vector-model curve is not directly transferable in the low-mass region where semi-Compton production dominates. Above $m_{A'} = 2m_\mu$, the axial and vector dimuon widths also differ in their threshold dependence, as quantified under Major Point 4.3, affecting visible-decay and low-energy-supernova energy-deposition constraints.

Kinetic-mixing-induced electromagnetic processes.

The low-momentum kinetic-mixing estimate has a separate scope. In the equal-mixing benchmark,

$$\frac{|\epsilon_{\gamma A'}(0)|}{|g_X|} \simeq 1.4 \times 10^{-6},$$

compared with approximately 1/70 in the conventional vector model. At the same vacuum kinematics, this suppresses only electromagnetic-current-mediated amplitudes and rates, such as plasmon conversion or the mixing-induced proton coupling in np bremsstrahlung, by about 10^{-4} and 10^{-8}, respectively. It does not suppress neutrino coalescence or the direct axial semi-Compton process. These factors are same-kinematics vacuum scaling estimates, not supernova emissivity ratios; an actual in-medium rate requires the momentum-dependent mixing and plasma propagators.

An explicit axial-EFT cooling benchmark. We evaluate the axial interaction at the effective-current level, with $g_A^\mu = -g_X$, $g_V^\mu = 0$, and the active-neutrino currents above. An identical ultraviolet completion is not required to reuse a calculation with these operators. Existing axial-vector cooling estimates use longitudinal-mode matching [38, 39]; here finite-mass semi-Compton scattering is obtained from the exact tree-level spin-one kernels of Liu and Miller [40], with the electron mass and coupling replaced by those of the muon. The small vector admixture described under Major Point 4.3 can be retained in this EFT; the plotted benchmark is its pure-axial limit.

For $\gamma(k)\mu(p) \to A'(q)\mu(p')$, define $M = m_{A'}$, $m = m_\mu$, $S = (p+k)^2 - m^2$, $U = (p-q)^2 - m^2$, and $t = (p'-p)^2$. Then

$$S + U + t = M^2, \quad R = \frac{(S+U)^2}{SU}, \quad B = m^2\left(\frac{S+U}{SU}\right)^2 - \frac{t}{SU},$$

and the initial-spin-averaged, final-polarization-summed result is

$$\overline{|\mathcal{M}|^2} = e^2\left[(g_V^\mu)^2\mathcal{A}_V + (g_A^\mu)^2\mathcal{A}_A\right], \quad \mathcal{A}_V = 4 - 2R + 4(M^2 + 2m^2)B,$$

$$\mathcal{A}_A = 4 - \left(2 + \frac{4m^2}{M^2}\right)R + 4(M^2 - 4m^2)B, \qquad \frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{16\pi S^2}.$$

The VA interference vanishes for this unpolarized vacuum $2 \to 2$ observable. These expressions retain the full masses and do not use the high-energy Weizsäcker–Williams approximation. An independent explicit-spinor evaluation agrees with the published kernels in 28 pure/mixed-current comparisons, with maximum relative difference 2.9×10^{-12}.

For transport we keep each mediator polarization $h = L, +, -$ separately. Inverse Compton scattering is evaluated with an initial muon at rest, exact two-body recoil, and the effective muon density

$$n_\mu^{\rm eff}(r) = 2\sum_{\eta=\pm1}\int \frac{d^3p}{(2\pi)^3} f_\mu(E;\eta\mu_\mu)\,[1 - f_\mu(E;\eta\mu_\mu)].$$

Writing $k_A = \sqrt{\omega^2 - M^2}$, $D = m + \omega - k_A c$, the outgoing photon energy and differential absorption kernel are

$$\omega_\gamma = \frac{M^2 + 2m\omega}{2D}, \qquad v_A \frac{d\sigma_h^{\rm abs}}{d\Omega} = \frac{\overline{|\mathcal{M}_h^{\rm abs}|^2}\,\omega_\gamma}{64\pi^2 m\omega D}.$$

Here only the initial muon spin is averaged. Multiplication by $n_\mu^{\rm eff}$, angular integration and the factor $1+f_B(\omega_\gamma)$ give $\Gamma_h^{\rm abs}$. Production is assigned through thermal detailed balance,

$$\Gamma_h^{\rm prod} = e^{-\omega/T}\Gamma_h^{\rm abs}, \qquad \Gamma_h^{\rm red} = (1 - e^{-\omega/T})\Gamma_h^{\rm abs}.$$

This is the stated muon-rest/effective-density approximation, with exact recoil; it is not an exact moving-muon thermal integral.

Neutrino production and decay are computed with Fermi–Dirac occupations and Pauli blocking. For each flavor, $\Gamma_{\nu,0} = g_X^2 M/(24\pi)$, and

$$\Gamma_{\nu,h}^{\rm prod/abs} = \frac{M}{\omega}\Gamma_{\nu,0}\int_{-1}^{1} dz\, W_h(z) \begin{cases} f_\nu(E_+;\mu_\nu) f_{\bar\nu}(E_-;-\mu_\nu), \\ [1 - f_\nu(E_+;\mu_\nu)][1 - f_{\bar\nu}(E_-;-\mu_\nu)], \end{cases}$$

$$E_\pm = \frac{\omega \pm k_A z}{2}, \quad W_L = \frac{3}{4}(1 - z^2), \quad W_\pm = \frac{3}{8}(1 \pm z)^2, \quad \int_{-1}^{1} W_h\, dz = 1.$$

The two flavors are summed with their respective chemical potentials, taking $\mu_{\nu_\tau} = 0$. In particular, a nonzero μ_{ν_μ} can distinguish the two circular transverse rates, so their opacities are not averaged before exponentiation. Above $M = 2m_\mu$, dimuon inverse decay and decay are included with the axial width $g_X^2 M(1 - 4m_\mu^2/M^2)^{3/2}/(12\pi)$, the appropriate massive two-body kinematics and muon blocking factors.

We use a reconstructed SFHo-18.8 profile at one second after bounce. The temperature and muon/neutrino chemical potentials are extracted from the published vector-graphics figure in Ref. [13]; the gravitational lapse is extracted from Ref. [41]. These are digitized published profiles, not the original simulation tables. Reconstructing the muon density from the extracted T, μ_μ

agrees with the independently plotted density at five radii between 0 and 15 km to approximately 0.5–1.1%. The two published temperature curves agree within 0.27% over their common range. The last kilometre of the lapse is continued by an exterior Schwarzschild form matched at 24 km; unresolved sub-MeV neutrino chemical potentials are set to zero.

Following the ballistic-cooling prescription of Ref. [26], the luminosity at infinity is

$$L_{A'} = \sum_h \int_0^{R_\nu} 4\pi r^2 dr\, N(r)^2 \int_{M/N(r)}^{\infty} d\omega\, \frac{k_A \omega^2}{2\pi^2}\, \Gamma_h^{\mathrm{prod}}(r,\omega)\, \mathcal{P}_h(r,\omega),$$

$$\mathcal{P}_h = \frac{1}{2}\int_{-1}^{1} dc_\theta\, \exp\!\left[-\int_0^{s_{\max}} \frac{ds}{v_A}\, \Gamma_h^{\mathrm{red}}\!\left(\sqrt{r^2+s^2+2rsc_\theta},\omega\right)\right],$$

$$s_{\max} = -rc_\theta + \sqrt{R_\nu^2 - r^2 + r^2 c_\theta^2}, \qquad R_\nu = 25\ \mathrm{km}.$$

This includes inward and outward trajectories, the velocity factor, stimulated-emission reduction of opacity, gravitational redshift of luminosity and the escape-energy threshold. As in this prescription, the local energy is held fixed along a straight chord; this is not full general-relativistic ray transport. The polarization sum is outside the escape exponential. The cooling criterion is $L_{A'} > 4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$, with the alternative $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ also shown.

Representative intervals above the $4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ criterion are

M [MeV]	$\lvert g_X\rvert_{\mathrm{lower}}$	$\lvert g_X\rvert_{\mathrm{upper}}$
1	3.8×10^{-11}	4.9×10^{-7}
10	3.6×10^{-10}	7.5×10^{-8}
100	2.4×10^{-10}	7.3×10^{-9}
200	4.1×10^{-10}	2.9×10^{-9}

and the benchmark closes near $M \simeq 0.26$ GeV. The lower edge is production limited; the upper edge includes absorption and decay trapping. At $M = 10$ MeV, semi-Compton emission supplies approximately 91% of the optically thin axial luminosity, whereas neutrino coalescence supplies approximately 99.4% at 100 MeV. Thus the vector benchmark's quoted dominance above approximately 10 MeV cannot be assumed unchanged for the axial current.

Several checks determine the precision of this comparison. The kernels reproduce the vector Thomson limit, the axial longitudinal mass scaling, the vacuum neutrino/dimuon widths and mode-specific detailed balance. The angular escape calculation reproduces the analytic uniform-sphere result.

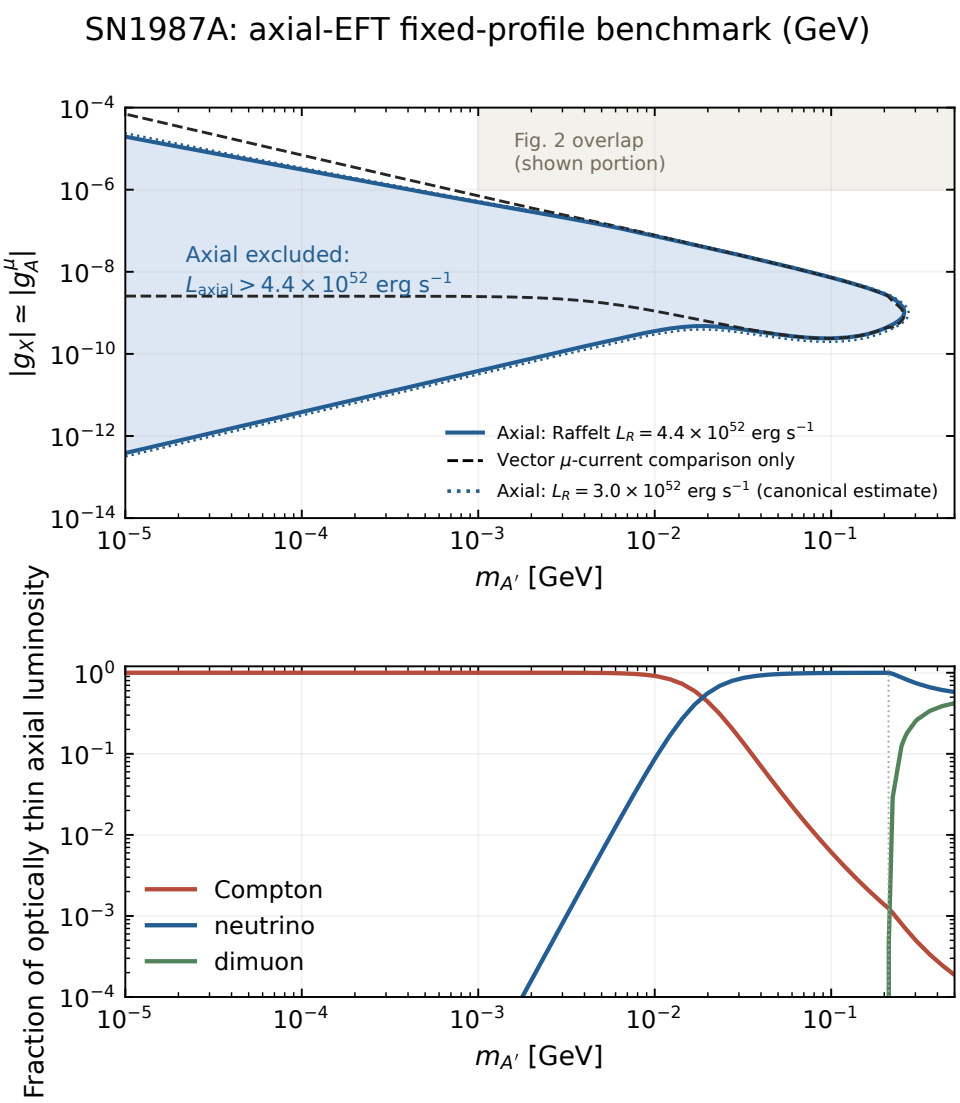


Figure 1: Conditional SN1987A ballistic-cooling exclusion for the axial EFT, with $m_{A'}$ shown in GeV for direct comparison with Fig. 2. The GeV mass axis is obtained by converting the same saved MeV scan, $m_{A'}[\mathrm{GeV}] = m_{A'}[\mathrm{MeV}]/1000$; no scan boundary or physical input is changed. Top: the blue shading marks the axial region with $L_{\rm axial} > 4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$. The solid blue boundary uses $L_R = 4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$, the one-second Raffelt luminosity adopted for the Garching supernova profiles in the recent $L_\mu - L_\tau$ analyses of Refs. [26, 27]; it is also the reference luminosity associated with the SFHo-18.8 benchmark used here. The dotted blue curve uses the alternative canonical $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ luminosity estimate obtained from the local emissivity criterion at fiducial core conditions. Thus 3.0×10^{52} is a threshold comparison, not the primary Raffelt luminosity adopted in the cited profile-based vector-boson studies. The black dashed curve is an internal vector-muon-current control, not a published exclusion contour. Although the vector current, active-neutrino sector and nominal transport prescription are matched within this code, the reconstructed thermodynamic profile and muon-rest/effective-density treatment have not yet been demonstrated to reproduce the published vector cooling contours point by point; the curve is therefore diagnostic only and is not used as an independently validated bound. The beige

Increasing the radial, energy and angular quadrature gives stable lower edges at the sub-percent level; the most sensitive low-mass upper edge changes by approximately 8% in the refinement checks, while the 1–250 MeV upper edges are substantially more stable. These numerical checks are separate from physical approximation uncertainties. Scaling the semi-Compton rate and its inverse together by factors 1/2 and 2 moves the low-mass lower edge by approximately +41% and –29%, respectively. This is a sensitivity test, not a confidence interval. Changing the common source/escape radius to 20 km changes the upper edge by approximately a factor of two at 10 keV and by 25% at 100 MeV; setting $\mu_{\nu_\mu} = 0$ changes the tested boundaries by less than 1%.

In the relativistic longitudinal limit the result can also be checked against the corrected ALP–muon bound of Bollig et al. [42, 43]:

$$\widehat{g}_{a\mu} = \frac{|g_A^\mu|}{M}, \qquad \widehat{g}_{a\mu} \lesssim 10^{-7.5}\ \mathrm{GeV}^{-1}.$$

For the $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ criterion our low-mass lower edge is approximately

$$|g_A^\mu|_{\mathrm{lower}} \simeq 3.2 \times 10^{-11}(M/\mathrm{MeV}),$$

consistent with that longitudinal estimate. The finite-mass calculation and its upper edge do not follow from extending this straight line.

The plotted exclusion is specifically a stationary-profile ballistic-cooling benchmark. Muon thermal motion and full finite-temperature photon dispersion are not included; muon–proton bremsstrahlung and other additional channels are not part of this three-channel estimate. The approximate rate rescaling above tests sensitivity without replacing a calculation of those effects. Sufficiently strong new interactions can also change the underlying supernova profile. The diffusive-transport, high-energy-neutrino and LESN observables discussed in Ref. [26] impose different criteria and are not represented by this shading. Within the stated benchmark the ballistic exclusion lies below the $g_X \geq 10^{-6}$, $M \geq 1$ MeV portion displayed in the original Fig. 2. This explains why that laboratory-summary window and the new cooling figure cover complementary couplings; it does not establish that astrophysical energy transport is irrelevant within the original window.

Interpretation and applicability.

The vector-model result provides a reference for the neutrino-coalescence channel in the cited profile for $m_{A'} \gtrsim 10\,\mathrm{MeV}$; the full cooling and transport assessment depends on the axial and in-medium processes discussed above.

Minor Point 1).

> *Range of parameters. The general construction of the model is clearly written, but it is difficult for the reader to envision the mass and coupling range that the authors are targeting. It would be very helpful if the authors decide on a range of $m_{A'}$ and g_X of interest. For example: is there a range of masses where experimental or theoretical limits are less constraining? Is there some set of phenomenology that the authors seek to realize with a "light" A'? If so, they should define "light." (See, e.g. the first word of the introduction, which is not defined with respect to any reference mass scale.)*

Reply. The word "light" is not intended to specify a separate quantitative mass regime. The Abstract describes the broader MeV-to-TeV scale, while Fig. 2 displays the specific comparison window

$$10^{-3}\,\mathrm{GeV} \leq m_{A'} \leq 10^{2}\,\mathrm{GeV}, \qquad 10^{-6} \leq g_X \leq 1.$$

The numerical range and the observables included in the analysis, rather than the adjective "light", define the phenomenological domain considered here. The lower end retains the low-mass region in which the laboratory $(g-2)_\mu$ constraint is already important; its asymptotic continuation and the overlapping, model-dependent astrophysical sensitivity are discussed under Major Points 2 and 4.4. At the upper end, the meson-decay, B-factory, and Z-decay searches included in Fig. 2 approach or exhaust their kinematic coverage, while heavier mediators require higher-energy probes. Neither endpoint therefore marks a physical boundary of the model or a universal exclusion boundary. The detailed motivation for the window and the complementary experimental coverage is given in our response to Major Point 2.

Minor Point 2).

> *Defining the pure-axial limit. Equation (12) gives the coupling of the A' to the mass-basis fields. It is here that the "pure axial coupling" limit is defined? i.e. $\theta_{\mu R} \to 0$? If that is the case, it would help to define this as the pure-axial limit, a phrase which is used below (36) but not otherwise quantified.*

Reply.

We agree that the pure-axial limit should be quantified when the mass-basis couplings are introduced in Eq. (12). In the convention used there, the strict pure-axial limit is $\theta_{\mu R} \to 0$:

$$g_V^\mu = g_X \sin^2\theta_{\mu R} \to 0, \qquad g_A^\mu = -g_X \cos^2\theta_{\mu R} \to -g_X.$$

For finite $\theta_{\mu R}$, the vector coupling is nonzero, so the interaction is not strictly purely axial. In the tau sector, the opposite $U(1)_X$ charge assignment gives the opposite overall sign, while the coupling magnitudes are $|g_V^\tau| = |g_X| \sin^2\theta_{\tau R}$ and $|g_A^\tau| = |g_X| \cos^2\theta_{\tau R}$. Thus $|g_V^\tau / g_A^\tau| = \tan^2\theta_{\tau R}$, and the strict pure-axial limit of the full charged-lepton current requires both right-handed mixing angles to vanish. Finite right-handed mixing angles describe a small vector admixture rather than the strict pure-axial limit.

Minor Point 3).

> *Bounds on the vectorlike leptons. Because the mixing angles scale inversely with the vectorlike lepton masses, it would be helpful to present lower limits on these masses. This, in turn, gives an upper limit on the $\theta_{(\mu,\tau)R}$ mixing angles which then gives an upper limit on the A' vectorlike coupling to $L_{\mu-\tau}$. How do these bounds compare with the $\theta_{(\mu,\tau)(L,R)}$ bounds in equation (27)?*

Reply.

In the decoupling regime, the light–heavy mixing angles scale inversely with the vector-like-lepton masses at fixed off-diagonal mass parameters. After matching to the observed charged-lepton masses, Eqs. (21) and (23) give

$$m_\ell \simeq \theta_{\ell L}\theta_{\ell R} M_{E\ell}, \qquad \ell = \mu, \tau.$$

Combining these relations with the small-mixing condition adopted in Eq. (27),

$$|\theta_{\ell L}|,\ |\theta_{\ell R}| \leq 10^{-2},$$

gives the following conditional mass scales at leading order:

$$M_{E\mu} \gtrsim \frac{m_\mu}{(10^{-2})^2} \simeq 1.06 \text{ TeV}, \qquad M_{E\tau} \gtrsim \frac{m_\tau}{(10^{-2})^2} \simeq 17.8 \text{ TeV}.$$

For the degenerate choice $M_{E\mu} = M_{E\tau} \equiv M_E$ in Eq. (22), the tau relation therefore gives the stronger conditional requirement $M_E \gtrsim 17.8$ TeV. This

follows from imposing the mixing-angle condition and reproducing the charged-lepton masses simultaneously; it is not an independent direct-search mass limit. The separate condition $M_E > 10$ TeV in Eq. (22) is weaker than this combined requirement.

Equation (27) and the mass relations play different roles: Eq. (27) bounds each mixing angle individually in the adopted small-mixing setup, whereas the mass relations fix the products $\theta_{\ell L}\theta_{\ell R}$. A mass relation by itself would not yield an individual upper bound on $\theta_{\ell R}$. Using the Eq. (27) condition in the mass-basis couplings,

$$
\begin{aligned}
&|g_V^\mu| = |g_X| \sin^2\theta_{\mu R}, \qquad |g_A^\mu| = |g_X| \cos^2\theta_{\mu R},\\
&|g_V^\tau| = |g_X| \sin^2\theta_{\tau R}, \qquad |g_A^\tau| = |g_X| \cos^2\theta_{\tau R},\\
&\left|\frac{g_V^\ell}{g_A^\ell}\right| = \tan^2\theta_{\ell R} \lesssim 1.01 \times 10^{-4}, \qquad \ell = \mu, \tau.
\end{aligned}
$$

The signed vector couplings have opposite overall signs in the muon and tau sectors because their $U(1)_X$ charges are opposite; their magnitudes obey the same bound, $|g_V^\ell| \lesssim 10^{-4}|g_X|$, for both $\ell = \mu, \tau$. The signed axial couplings likewise have opposite overall signs, while their absolute magnitudes remain close to $|g_X|$ in this small-mixing domain. Thus, the residual vector component is suppressed by at least four orders of magnitude relative to the axial component under the adopted condition. This suppression follows from the adopted small-mixing condition rather than a separate experimental bound; its impact on the observables is quantified under Major Point 4.3.

Minor Point 4).

> *Conservative benchmark. Equation (27) is presented as a "conservative benchmark." What does this mean? A benchmark point would be a choice of these mixing angles. A constraint is an upper bound. Eq. (27) is written as an upper bound, but is described as a benchmark. This is not simply semantic: readers would want to understand if* 10^{-2} *is an experimental upper limit or simply a convenient choice.*

Reply. Equation (27) specifies the upper envelope adopted for the small-mixing domain; 10^{-2} is a deliberately chosen analysis assumption, not a single benchmark point or an experimental or global-fit upper limit. We impose this numerical bound separately on $\theta_{\ell L}$ and $\theta_{\ell R}$ for each $\ell = \mu, \tau$, without identifying the left- and right-handed angles or requiring the muon and tau

angles to coincide. Together, these independent bounds define the near-axial working domain used in the analysis. At its upper edge, the leading mixing corrections to the weak and gauge currents scale as $\theta^2 \lesssim 10^{-4}$, while the right-handed rotation gives $|g_V^\ell/g_A^\ell| = \tan^2\theta_{\ell R} \lesssim 1.01 \times 10^{-4}$. For comparison, the LEP leptonic-width constraints quoted in the model discussion apply to left-handed muon mixing at the few-percent level [17, 44]; they therefore provide context for $\theta_{\mu L}$, not a direct bound on all four angles. Other vector-like-lepton fits depend on the electroweak representation and the observables included in the fit [45, 46]. The corresponding conditional VLL mass scales follow from the charged-lepton mass relations discussed under Minor Point 3.

Minor Point 5).

> *What is the purpose of including muon collider bounds? The second half of the paper focuses on existing experimental bounds on the effective theory. It is a strange change in tone to include hypothetical muon collider bounds for future reach. The authors do not justify why a future muon collider is the way to test this model in the future and the section feels like it is tacked on. Most of the section emphasizes that the muon collider search would be a "standard" follow up of existing work (Ref 83) and so this section does not seem particularly novel. The authors simply focus on the forward-backward asymmetry. Why not also include polarization asymmetries? Or would there be asymmetries in the ee versus $\tau\tau$ final states? Given the first half of the paper, though, it seems that the real "smoking gun" for this UV completion is the physics of the S particle which should be tested in any Higgs-factory phase of a future lepton collider. I feel that this section either adds too little and should be downplayed as an example of future tests, or otherwise should have its own full fledged exploration (which may be a separate paper).*

Reply. A high-energy muon collider is a natural complementary probe because A' couples directly to the initial-state muons and can therefore be tested in the heavier-mass regime, beyond the low-energy probes considered in the phenomenological analysis. Ref. [47] provides a useful vector $L_\mu - L_\tau$ collider benchmark at $\sqrt{s} = 3\,\text{TeV}$. The axial model, however, cannot be assessed by simply importing the vector result: the left- and right-handed couplings entering the helicity amplitudes are different.

We use the $\tau^+\tau^-$ final state because it receives only s-channel γ, Z, A'

exchange; the same-flavor $\mu^+\mu^-$ final state also contains t-channel diagrams, which complicate the angular interpretation.

For the pure-axial, heavy-VLL benchmark with $m_{A'} \gg m_\tau$, the tree-level partial widths are $\Gamma(A' \to \ell^+\ell^-) = g_X^2 m_{A'}/(12\pi)$ for each charged-lepton flavor $\ell = \mu, \tau$, and $\Gamma(A' \to \nu_\ell \bar{\nu}_\ell) = g_X^2 m_{A'}/(24\pi)$ for each active chiral-neutrino flavor $\ell = \mu, \tau$. When these four channels are open, the heavy VLL pair channels are kinematically closed, and the loop-induced electron mode is neglected, the total width is $\Gamma_{A'} = g_X^2 m_{A'}/(4\pi)$. Thus, the nonresonant $m_{A'} = 1\,\text{TeV}$, $g_X = 0.05$ point in the left panel has $\Gamma_{A'} \simeq 0.199\,\text{GeV}$, whereas the quoted $0.60\,\text{GeV}$ width corresponds to the resonance point $m_{A'} = \sqrt{s} = 3\,\text{TeV}$ in the right-panel scan. This total width enters the A' propagator in the chiral amplitudes below. For $\mu^+\mu^- \to \tau^+\tau^-$, neglecting the external-lepton masses, the unpolarized angular distribution is

$$\frac{d\sigma_{\tau\tau}}{d\cos\theta} = \frac{s}{128\pi}\Big[\left(|\mathcal{A}_{LL}|^2 + |\mathcal{A}_{RR}|^2\right)(1+\cos\theta)^2 + \left(|\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2\right)(1-\cos\theta)^2\Big],$$

where the chiral amplitudes are

$$\mathcal{A}_{ij} = \frac{e^2}{s} + \frac{g_i^Z g_j^Z}{s - m_Z^2 + i m_Z \Gamma_Z} + \frac{g_i^\mu g_j^\tau}{s - m_{A'}^2 + i m_{A'}\Gamma_{A'}}, \qquad i,j = L,R,$$

and the Standard Model Z-boson couplings are

$$g_Z = \frac{e}{s_W c_W}, \qquad g_L^Z = g_Z\left(-\frac{1}{2} + s_W^2\right), \qquad g_R^Z = g_Z s_W^2.$$

Here i labels the initial-state muon chirality and j the final-state tau chirality. We define θ as the angle between the incoming μ^- and outgoing τ^- momenta in the center-of-mass frame, so $\cos\theta > 0$ is the forward τ^- hemisphere. The vector and axial charge assignments are

$$\begin{aligned} \text{vector:} \quad & (g_L^\mu, g_R^\mu, g_L^\tau, g_R^\tau) = (g_X, g_X, -g_X, -g_X), \\ \text{axial:} \quad & (g_L^\mu, g_R^\mu, g_L^\tau, g_R^\tau) = (g_X, -g_X, -g_X, g_X). \end{aligned}$$

The sign reversal between the left- and right-handed couplings changes the helicity amplitudes and the relative weights of the $(1+\cos\theta)^2$ and $(1-\cos\theta)^2$ terms. Thus a total-rate reinterpretation of the vector result can obscure the axial-current effect, whereas the angular distribution retains sensitivity to the chiral structure.

For the same central angular acceptance $|\cos\theta| < 0.9$ used in the figure, the cut forward–backward asymmetry is

$$A_{\rm FB} = \frac{\displaystyle\int_0^{0.9} d\cos\theta\,\frac{d\sigma}{d\cos\theta} - \int_{-0.9}^{0} d\cos\theta\,\frac{d\sigma}{d\cos\theta}}{\displaystyle\int_{-0.9}^{0.9} d\cos\theta\,\frac{d\sigma}{d\cos\theta}}$$

The asymmetry isolates the forward–backward odd part of the distribution and is therefore a chirality-sensitive observable that does not require polarized initial beams. The fully inclusive definition corresponds to replacing the cut value 0.9 by 1. For the collider benchmark $\sqrt{s} = 3\,\text{TeV}$, $m_{A'} = 1\,\text{TeV}$, and $g_X = 0.05$, the vector benchmark changes the total rate relative to the SM prediction in $|\cos\theta| < 0.9$ by about 5%, whereas the axial benchmark changes it by about 2%. The axial benchmark gives $\Delta A_{\rm FB} \simeq -2.9 \times 10^{-2}$, larger in magnitude than the vector shift for the same benchmark. This comparison demonstrates why an angular observable is more discriminating than a simple total-rate rescaling for the axial interaction.

Writing

$$S_+ = |\mathcal{A}_{LL}|^2 + |\mathcal{A}_{RR}|^2, \qquad S_- = |\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2,$$

the angular integration gives

$$A_{\rm FB}(c_{\rm max}) = \frac{c_{\rm max}}{1 + c_{\rm max}^2/3}\,\frac{S_+ - S_-}{S_+ + S_-}.$$

For the plotted cut $c_{\rm max} = 0.9$, the purely kinematic prefactor is approximately 0.709. Thus $A_{\rm FB}$ directly compares the same-chirality and opposite-chirality helicity combinations.

Polarization observables would provide complementary information because they project separately onto initial- and final-state chiral amplitudes. Following the convention used in Ref. [47], an initial-state left–right asymmetry can be defined with a polarized μ^- beam and an unpolarized μ^+ beam:

$$A_{\rm LR}^{\mu} = \frac{\sigma(\mu_L^-\mu^+ \to \tau^+\tau^-) - \sigma(\mu_R^-\mu^+ \to \tau^+\tau^-)}{\sigma(\mu_L^-\mu^+ \to \tau^+\tau^-) + \sigma(\mu_R^-\mu^+ \to \tau^+\tau^-)}.$$

With unpolarized initial beams, final-state tau polarimetry can instead measure

$$P_\tau = \frac{\sigma_{\tau_R} - \sigma_{\tau_L}}{\sigma_{\tau_R} + \sigma_{\tau_L}},$$

where

$$\sigma_{\tau_R} \equiv \sigma(\mu^+\mu^- \to \tau_R^-\tau_L^+), \qquad \sigma_{\tau_L} \equiv \sigma(\mu^+\mu^- \to \tau_L^-\tau_R^+)$$

in the relativistic limit. For the same symmetric angular acceptance $|\cos\theta| < c_{\max}$ and in the massless-lepton approximation used above, the common angular integrals cancel in the ratios. With the first index of $\mathcal{A}_{ij}$ denoting the initial μ^- chirality and the second the final τ^- chirality, these definitions become

$$A_{\rm LR}^{\mu} = \frac{|\mathcal{A}_{LL}|^2 + |\mathcal{A}_{LR}|^2 - |\mathcal{A}_{RL}|^2 - |\mathcal{A}_{RR}|^2}{|\mathcal{A}_{LL}|^2 + |\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2 + |\mathcal{A}_{RR}|^2},$$

$$P_\tau = \frac{|\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RR}|^2 - |\mathcal{A}_{LL}|^2 - |\mathcal{A}_{RL}|^2}{|\mathcal{A}_{LL}|^2 + |\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2 + |\mathcal{A}_{RR}|^2}.$$

The first expression separates the two initial-muon chiralities, while the second separates the final tau helicities. They therefore supply information complementary to $A_{\rm FB}$, which instead compares same- and opposite-chirality amplitude combinations after summing over both beams and final-state helicities. We retain $A_{\rm FB}$ as the representative observable for the unpolarized-beam illustration; numerical sensitivity for the complementary polarization observables requires beam-polarization or tau-polarimetry efficiencies, analyzing powers, and systematic uncertainties.

The ee final state is not an equivalent tree-level channel in the pure-axial effective-current limit, because A' has no tree-level coupling to electrons. An electron coupling can arise through loop-induced kinetic mixing. The equal-mixing benchmark gives $|\epsilon_{\gamma A'}| \leq 2.12 \times 10^{-7}$ for $g_X \leq 0.1$ in the tabulated low-mass, on-shell range, corresponding to $e|\epsilon_{\gamma A'}| \lesssim 6.4 \times 10^{-8}$. This number is not a prediction for a multi-TeV muon collider: the mixing is momentum dependent, and the relevant virtuality for $\mu^+\mu^- \to e^+e^-$ is $q^2 = s$. A collider-level ee prediction would require evaluating $\epsilon_{\gamma A'}(s)$ with the corresponding matching condition. At the amplitude level, the direct $\tau^+\tau^-$ channel contains the tree-level product $g_i^\mu g_j^\tau$, whereas the ee channel contains an electron vertex proportional to $e\epsilon_{\gamma A'}(s)$. Its rate and any associated asymmetry therefore depend on the high-energy, momentum-dependent mixing, not on the low-mass on-shell number quoted under Major Point 4.2.

We agree that S is an important complementary UV target for a future Higgs factory. The renormalizable portal operator $(H^\dagger H)(S^\dagger S)$ is allowed by the gauge symmetry; a nonzero renormalized portal coefficient induces Higgs–singlet mixing. For $m_s < \sqrt{s_{ee}} - m_Z$, this mixing permits associated production $e^+e^- \to Zs$. In the single-doublet/singlet mixing description, the

tree-level rate is related to the Standard Model Higgsstrahlung rate by

$$\sigma(e^+e^- \to Zs) = \sin^2\theta_{hs}\,\sigma_{\rm SM}(e^+e^- \to ZH)\big|_{m_H=m_s},$$

so the production rate is controlled by the doublet component of s and the available phase space. Separately, if $m_s < m_h/2$, the exotic decay $h \to ss$ can be kinematically open. Its rate is controlled by the trilinear coupling g_{hss}, defined by $\mathcal{L} \supset -\frac{1}{2}g_{hss}hs^2$, and is not fixed by $\sin^2\theta_{hs}$ alone:

$$\Gamma(h \to ss) = \frac{g_{hss}^2}{32\pi m_h}\sqrt{1-\frac{4m_s^2}{m_h^2}}.$$

Both g_{hss} and θ_{hs} depend on the scalar potential and portal parameter. For the same small-portal matching benchmark specified under Major Point 1.3, the negligible-mixing description applies away from the narrow near-degenerate region identified there; that region is not covered by the small-mixing approximation. We therefore identify $e^+e^- \to Zs$ and $h \to ss$ as complementary channels whose rates depend on these scalar-potential parameters. The scalar decay pattern, including $s \to A'A'$ when allowed, further determines the visible final states. The Higgs-factory scalar programme is therefore complementary to, rather than replaced by, the muon-collider angular test of the A' current.

Accordingly, the muon-collider result is a parton-level example of a future chirality-sensitive test. Its purpose is to show that the axial interaction has a distinctive angular signature that is not captured by a vector-to-axial total-rate reinterpretation, while the scalar S remains a separate target for a future Higgs factory.

Minor Point 6).

Terminology. The following points could be improved for clarity.

Minor Point 6.1).

The word "light" on page 2 has no reference scale.

Reply. The adjective "light" is not used as a quantitative mass classification. The Abstract states the broader MeV-to-TeV range, while the explicit comparison window in Fig. 2 is

$$10^{-3}\,\text{GeV} \le m_{A'} \le 10^{2}\,\text{GeV}, \qquad 10^{-6} \le g_X \le 1.$$

These numerical ranges, together with the observables included in the analysis, define the phenomenological domain; the word "light" does not introduce an additional universal mass threshold. This distinction matters because the analysis combines probes with different kinematic thresholds and sensitivities over several orders of magnitude; no single mass cut separates all of them into a common *light* and *heavy* regime.

Minor Point 6.2).

> *At the bottom of page 11, when referring to Table II, the authors write "representative bounds." What does representative mean? Does it mean these are the most stringent bounds? Are these "some" bounds? Is the list "complete" in any sense?*

Reply. Here, "representative" means that Table II summarizes selected constraints from the literature that illustrate the different scalar-mass regimes and relevant physical mechanisms. The quoted values retain the coupling normalizations, mass ranges, and source assumptions of the corresponding analyses. The table is intended to provide an overview of relevant constraints, rather than an exhaustive survey or a ranking of the strongest bound at every mass.

Minor Point 6.3).

> *Between page 12–13: "...do not provide a universal g_X bound, but they show that the intermediate scalar-mass region cannot simply be assumed unconstrained." What does intermediate mass mean? It is also not clear what the intended content of this statement is: are the authors presenting a bound? Why would the reader assume that a new particle in this range would be unconstrained?*

Reply. The intermediate-mass range refers specifically to the direct BaBar and Belle searches for

$$e^+e^- \to \tau^+\tau^- s, \qquad s \to e^+e^- \text{ or } \mu^+\mu^-,$$

which cover approximately $0.04\,\mathrm{GeV} \lesssim m_s \lesssim 6.5$–$7\,\mathrm{GeV}$ [1, 2]. These searches provide direct experimental coverage of the intermediate-mass scalar region. For the present minimal UV benchmark with negligible Higgs–scalar mixing, the radial scalar has tree-level Yukawa couplings $y_{s\ell\ell} = m_\ell/v_s$ for $\ell = \mu, \tau$,

but no direct tree-level Yukawa coupling to electrons. The electron coupling induced by Higgs–scalar mixing, together with loop corrections to it, is neglected in this comparison. Accordingly, the UV translation uses the dimuon channel above threshold; the generic low-mass $s \to e^+e^-$ search channel is not treated as a direct constraint on this benchmark. In the mass-proportional benchmark, the Belle limit $\xi^{\rm lim} = 0.483$ at $m_s = 1\,\text{GeV}$ [2] gives

$$v_s > \frac{v_h}{0.483} \simeq 509\,\text{GeV}, \qquad y_{s\mu\mu} < 2.1 \times 10^{-4},$$

and, using $m_{A'} = 2g_X v_s$,

$$g_X < 9.8 \times 10^{-4} \left(\frac{m_{A'}}{\text{GeV}}\right)$$

at 90% confidence level for the illustrative $m_s = 1\,\text{GeV}$ point with $m_{A'} > 0.5\,\text{GeV}$, where $s \to A'A'$ is closed. This is separate from the $m_s \gtrsim 10\,\text{GeV}$ reference benchmark used for Fig. 2 and is not a universal bound on g_X; for $m_{A'} < 0.5\,\text{GeV}$, the visible branching fraction must be treated as described under Major Point 1.

The Belle II references cover two distinct analyses and should not be described collectively as spin-one-only searches. The $e^+e^- \to \mu^+\mu^- X$, $X \to \tau^+\tau^-$ analysis covers 3.6–10 GeV and explicitly includes a mass-proportional spin-zero leptophilic scalar [3]; the four-muon analysis covers 0.212–9 GeV and also tests a muonphilic scalar [4]. The former has the same charged-lepton Yukawa scaling as the present radial scalar, while the latter is a complementary muonphilic benchmark. For the UV completion considered here, the additional $s \to A'A'$ width modifies the visible dilepton or ditau branching fractions when the channel is open. The Belle II scalar results therefore provide relevant published scalar sensitivity, but their coupling and $\sigma \times \text{BR}$ limits translate into the present UV parameters only after using the model-specific total width and visible branching fractions, as detailed under Major Point 1.

Minor Point 6.4).

> *Below (34): "we impose Eq. (34) point by point", it's not clear what is meant by "point."*

Reply. A "point" denotes one grid point $(m_{A'}, g_X)$ in the parameter scan shown in Fig. 2. At each point, the symmetry-breaking scale is derived from

$$v_s = \frac{m_{A'}}{2g_X},$$

rather than varied independently. For a specified scalar mass m_s, the scalar self-coupling is then determined by

$$\lambda_S = \frac{m_s^2}{2v_s^2} = \frac{2g_X^2 m_s^2}{m_{A'}^2}.$$

The condition

$$\lambda_S \leq \lambda_S^{\max}$$

is consequently checked at each scan point for the adopted scalar benchmark. Equivalently, the dimensionless ratio

$$\frac{\lambda_S}{4\pi} = \frac{m_s^2}{8\pi v_s^2}$$

must not exceed unity when the illustrative choice $\lambda_S^{\max} = 4\pi$ is used. This makes explicit that changing $m_{A'}$ or g_X changes the derived v_s, and hence changes the scalar self-coupling tested at that grid point.

Equation (34) is checked point by point to test the theoretical consistency of the chosen scalar benchmark; it is not itself an experimental exclusion. The experimental constraints are a separate step.

Minor Point 6.5).

> *Page 21, first paragraph of Section E. What is the "parameter region of interest"? This wasn't clearly defined anywhere.*

Reply. We agree that the phrase "parameter region of interest" did not specify the mass coverage of the searches considered in Section III.E. The relevant four-muon searches cover approximately

$$\begin{aligned} \text{BaBar}: &\quad 0.212\text{–}10\,\text{GeV}, \\ \text{Belle II}: &\quad 0.212\text{–}9\,\text{GeV}, \\ \text{CMS}: &\quad 5\text{–}70\,\text{GeV}. \end{aligned}$$

These intervals describe the mass coverage of the published analyses; they do not imply that the entire intervals are excluded in our model.

The relevant processes are

$$e^+e^- \to \mu^+\mu^- A', \qquad A' \to \mu^+\mu^-,$$

and

$$Z \to \mu^+\mu^- A', \qquad A' \to \mu^+\mu^-.$$

The reinterpretation accounts for the production rate and $\mathrm{Br}(A' \to \mu^+\mu^-)$. The experimental analyses also impose prompt-resonance selection requirements. Their limits are applicable to our model only where the predicted A' decay length is compatible with those requirements; the mass coverage alone does not establish this condition throughout the parameter space.

In the original one-VEV model, the massless active neutrinos couple through

$$\mathcal{L} \supset g_X A'_\alpha \left(\bar{\nu}_\mu \gamma^\alpha P_L \nu_\mu - \bar{\nu}_\tau \gamma^\alpha P_L \nu_\tau\right).$$

A single massless chiral-neutrino channel contributes $g_X^2 m_{A'}/(24\pi)$; summing the two active flavors therefore gives

$$\Gamma_{\nu\bar{\nu}} = \frac{g_X^2 m_{A'}}{12\pi}.$$

Because this is a positive partial width, $\Gamma_{\mathrm{tot}} \geq \Gamma_{\nu\bar{\nu}}$, and hence it gives an upper bound on the proper decay length:

$$c\tau_{A'} = \frac{\hbar c}{\Gamma_{\mathrm{tot}}} \leq \frac{\hbar c}{\Gamma_{\nu\bar{\nu}}}.$$

At $m_{A'} = 0.212\,\mathrm{GeV}$ and $g_X = 7 \times 10^{-4}$, this yields $\Gamma_{\nu\bar{\nu}} = 2.76 \times 10^{-9}\,\mathrm{GeV}$ and $c\tau_{A'} \leq 0.0716\,\mu\mathrm{m}$. For a representative boost $\gamma \simeq 25$, the corresponding laboratory mean flight distance is at most $1.79\,\mu\mathrm{m}$. This is the quantity relevant to the prompt-resonance selection; the estimate is a local benchmark check, not a statement about all masses and couplings.

The benchmark lies only about $0.683\,\mathrm{MeV}$ above the dimuon threshold, since $2m_\mu \simeq 0.211317\,\mathrm{GeV}$. The axial dimuon width contains the threshold factor

$$\Gamma(A' \to \mu^+\mu^-) \simeq \frac{g_X^2 m_{A'}}{12\pi}\left(1 - \frac{4m_\mu^2}{m_{A'}^2}\right)^{3/2},$$

in the pure-axial limit. At this point the factor is approximately 5.16×10^{-4}, so the visible dimuon channel is strongly phase-space suppressed. The neutrino-only estimate is consequently a conservative lifetime upper bound near threshold; any additional open channel can only shorten the lifetime. The actual prompt requirement remains analysis-dependent.

The vector-to-axial rate-matching approximation is controlled for $m_{A'} \gg m_\mu$.

Near the dimuon threshold, a rate-level comparison alone is insufficient to infer a detector-level exclusion.

Minor Point 6.6).

> *Page 21, end of section D: What's the significance of the* $v_s = 2\,\mathrm{GeV}$ *line? What does it mean that* $y_{s\mu\mu}$ *"becomes large" (non-perturbative)? Why plot it? Is it a perturbativity bound?*

Reply. The line at $v_s \simeq 2\,\mathrm{GeV}$ follows from the scalar self-coupling condition for the reference benchmark $m_s = 10\,\mathrm{GeV}$ and $\lambda_S^{\max} = 4\pi$:

$$v_s \geq \frac{m_s}{\sqrt{2\lambda_S^{\max}}} = \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}.$$

At this reference value, $y_{s\mu\mu} = m_\mu/v_s \simeq 0.053$, so $y_{s\mu\mu}^2/(4\pi) \simeq 2.2 \times 10^{-4} \ll 1$. More generally, for the adopted $\lambda_S^{\max} = 4\pi$ criterion the boundary scales as

$$v_s^{\min}(m_s) = \frac{m_s}{\sqrt{8\pi}} \simeq 0.1995\, m_s.$$

Thus the plotted $v_s \simeq 2\,\mathrm{GeV}$ line corresponds specifically to the reference value $m_s = 10\,\mathrm{GeV}$; choosing a different scalar mass moves the boundary linearly with m_s. The line is shown to identify where that reference scalar benchmark first satisfies the scalar-quartic perturbativity condition. It is therefore a scalar-quartic consistency boundary for the stated m_s benchmark, not a Yukawa nonperturbativity limit or an experimental exclusion. Its relation to the point-by-point scalar-mass condition is given in Major Point 1.2 and Major Point 2.

Summary:

We construct an anomaly-free and renormalizable axial $U_A(1)_{L_{\mu-\tau}}$ completion in which the singlet scalar S and vector-like leptons generate the charged-lepton masses through a universal-seesaw mechanism. For the negligible-portal one-VEV benchmark, the unmixed tree-level radial-scalar mass parameter and the charged-lepton Yukawa couplings obey $m_s = \sqrt{2\lambda_S}\, v_s$ and $y_{s\ell\ell} \simeq m_\ell/v_s$, respectively. Thus, at fixed $(m_{A'}, g_X)$, the scalar–lepton couplings are fixed by $v_s = m_{A'}/(2g_X)$, whereas m_s is controlled by λ_S. The radial scalar remains a physical UV degree of freedom; its collider and low-mass phenomenology depends on its production rate, visible branching fractions, mass, and Higgs–scalar mixing. Table II summarizes selected low-mass probes

together with the coupling normalizations and source assumptions used in the corresponding analyses. For entries that directly constrain the scalar–muon Yukawa normalization, the one-VEV relation gives the conditional mapping $g_X \lesssim y^{\rm lim}_{s\mu\mu}(m_s) m_{A'}/(2m_\mu)$ at the corresponding m_s. The fifth-force entry instead constrains a loop-induced scalar–proton coupling, while the photon-mediated supernova entries use the muon-only loop and production treatment of the cited source. Their UV interpretation therefore retains the corresponding loop-matching and source assumptions.

In the original S-only symmetry-breaking limit, the residual Z_2 symmetry separates the electron flavor from the (μ, τ) sector. The resulting PMNS matrix permits a μ–τ rotation but no electron-flavor admixture, and therefore cannot reproduce the observed solar and reactor mixing angles. The optional $\sigma + 3N_R$ type-I seesaw completion removes this texture restriction by generating a general Majorana mass matrix and can accommodate the observed three-flavor mixing, subject to the seesaw, perturbativity, heavy-state mass, and flavor conditions. This extension is confined to Major Points 3 and 4.1; the non-neutrino phenomenology retains the original one-VEV model.

The axial current shifts C_A in neutrino trident production and gives a longitudinally enhanced negative contribution to $(g-2)_\mu$. Finite light–heavy mixing generates a small vector admixture and momentum-dependent kinetic mixing; their leading effects are quantified for the adopted small-mixing benchmark. The finite vector component gives subleading contributions to neutrino trident production and $(g-2)_\mu$ in this regime. Separately, the vector-only and axial-only coupling reaches in the high-mass four-muon searches are of the same order away from the dimuon threshold.

For stellar cooling, the neutrino couplings and invisible partial widths retain the conventional $L_\mu - L_\tau$ normalization, and neutrino-coalescence production dominates in the cited profile for $m_{A'} \gtrsim 10\,\mathrm{MeV}$. Charged-muon semi-Compton production and inverse absorption instead probe the non-conserved axial current, including its longitudinal contribution, whereas electromagnetic-current-mediated processes depend on the loop-induced, momentum-dependent kinetic mixing. The vector-model supernova result therefore provides a reference for the neutrino-coalescence channel. These process differences distinguish the axial transport problem from the conventional vector benchmark. At a future muon collider, the angular distribution and forward–backward asymmetry in $\mu^+\mu^- \to \tau^+\tau^-$ provide complementary sensitivity to the chiral structure beyond a total-rate comparison. The separately evaluated axial-EFT ballistic-cooling curve in Fig. 1 is a conditional

benchmark under the stated criterion, not a complete SN1987A transport or exclusion result.

Sincerely yours,

Rundong Fang, Jinhui Guo, Ming Li, Jia Liu, Xiao-Ping Wang, Yiheng Xiong

References

[1] **BaBar** Collaboration, J. P. Lees et al., *Search for a Dark Leptophilic Scalar in e^+e^- Collisions*, *Phys. Rev. Lett.* **125** (2020), no. 18 181801, [arXiv:2005.01885].

[2] **Belle** Collaboration, D. Biswas et al., *Search for a dark leptophilic scalar produced in association with $\tau^+\tau^-$ pair in e^+e^- annihilation at center-of-mass energies near 10.58 GeV*, *Phys. Rev. D* **109** (2024), no. 3 032002, [arXiv:2207.07476].

[3] **Belle-II** Collaboration, I. Adachi et al., *Search for a $\tau^+\tau^-$ resonance in $e^+e^- \to \mu^+\mu^-\tau^+\tau^-$ events with the Belle II experiment*, *Phys. Rev. Lett.* **131** (2023), no. 12 121802, [arXiv:2306.12294].

[4] **Belle-II** Collaboration, I. Adachi et al., *Search for a μ+μ- resonance in four-muon final states at Belle II*, *Phys. Rev. D* **109** (2024), no. 11 112015, [arXiv:2403.02841].

[5] T. Robens and T. Stefaniak, *Status of the Higgs Singlet Extension of the Standard Model after LHC Run 1*, *Eur. Phys. J. C* **75** (2015) 104, [arXiv:1501.02234].

[6] T. Robens, T. Stefaniak, and J. Wittbrodt, *Two-real-scalar-singlet extension of the SM: LHC phenomenology and benchmark scenarios*, *Eur. Phys. J. C* **80** (2020) 151, [arXiv:1908.08554].

[7] I. Boiarska, K. Bondarenko, A. Boyarsky, V. Gorkavenko, M. Ovchynnikov, and A. Sokolenko, *Phenomenology of GeV-scale scalar portal*, *JHEP* **11** (2019) 162, [arXiv:1904.10447].

[8] Z. Liu and Z.-W. Tang, *Probing a muonic force with the periastron advance in binary pulsar systems*, *Phys. Rev. D* **112** (2025), no. 5 055021, [arXiv:2501.10927].

[9] D. F. G. Fiorillo, A. Lella, G. G. Raffelt, N. Selimovic, and E. Vitagliano, *Neutron Star Bounds on Muonic Fifth Forces from Picometer to Kilometer Scales*, arXiv:2605.24094.

[10] J. K. Hoskins, R. D. Newman, R. Spero, and J. Schultz, *Experimental tests of the gravitational inverse square law for mass separations from 2-cm to 105-cm*, *Phys. Rev. D* **32** (1985) 3084–3095.

[11] D. J. Kapner, T. S. Cook, E. G. Adelberger, J. H. Gundlach, B. R. Heckel, C. D. Hoyle, and H. E. Swanson, *Tests of the gravitational inverse-square law below the dark-energy length scale*, *Phys. Rev. Lett.* **98** (2007) 021101, [hep-ph/0611184].

[12] C. P. Blakemore, A. Fieguth, A. Kawasaki, N. Priel, D. Martin, A. D. Rider, Q. Wang, and G. Gratta, *Search for non-Newtonian interactions at micrometer scale with a levitated test mass*, *Phys. Rev. D* **104** (2021), no. 6 L061101, [arXiv:2102.06848].

[13] A. Caputo, G. Raffelt, and E. Vitagliano, *Muonic boson limits: Supernova redux*, *Phys. Rev. D* **105** (2022), no. 3 035022, [arXiv:2109.03244].

[14] J. A. Dror, R. Laha, and T. Opferkuch, *Probing muonic forces with neutron star binaries*, *Phys. Rev. D* **102** (2020), no. 2 023005, [arXiv:1909.12845].

[15] S. Coleman and E. Weinberg, *Radiative corrections as the origin of spontaneous symmetry breaking*, *Phys. Rev. D* **7** (1973) 1888–1910.

[16] R. Jackiw, *Functional evaluation of the effective potential*, *Phys. Rev. D* **9** (1974) 1686–1701.

[17] **Particle Data Group** Collaboration, S. Navas et al., *Review of particle physics*, *Phys. Rev. D* **110** (2024), no. 3 030001.

[18] **PIENU** Collaboration, A. Aguilar-Arevalo et al., *Search for three body pion decays $\pi^+ \to l^+ \nu X$*, *Phys. Rev. D* **103** (2021), no. 5 052006, [arXiv:2101.07381].

[19] W. Altmannshofer, J. A. Dror, P. Giffin, S. Gori, O. Jackson, K. Luong, P. Schwendimann, and S. R. Seo, *Exploring Invisible New Physics with Exotic Pion Decays*, arXiv:2601.06254.

[20] **NA62** Collaboration, E. Cortina Gil et al., *Search for K^+ decays to a muon and invisible particles*, *Phys. Lett. B* **816** (2021) 136259, [arXiv:2101.12304].

[21] **CCFR** Collaboration, S. R. Mishra et al., *Neutrino Tridents and W Z Interference*, *Phys. Rev. Lett.* **66** (1991) 3117–3120.

[22] D. Geiregat, G. Wilquet, U. Binder, H. Burkard, U. Dore, W. Flegel, H. Grote, T. Mouthuy, H. Øverås, J. Panman, et al., *First observation of neutrino trident production*, *Physics Letters B* **245** (1990), no. 2 271–275.

[23] **NA64** Collaboration, Y. M. Andreev et al., *Shedding light on dark sectors with high-energy muons at the NA64 experiment at the CERN SPS*, *Phys. Rev. D* **110** (2024), no. 11 112015, [arXiv:2409.10128].

[24] **BaBar** Collaboration, J. P. Lees et al., *Search for a muonic dark force at BABAR*, *Phys. Rev. D* **94** (2016), no. 1 011102, [arXiv:1606.03501].

[25] **CMS** Collaboration, A. M. Sirunyan et al., *Search for an $L_\mu - L_\tau$ gauge boson using $Z\to 4\mu$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Lett. B* **792** (2019) 345–368, [arXiv:1808.03684].

[26] N. Blinov, P. J. Fox, K. J. Kelly, R. Plestid, and T. Zhou, *$L_\mu - L_\tau$ gauge bosons in beam dumps and supernovae*, *JHEP* **06** (2026) 155, [arXiv:2511.09619].

[27] Z. Wang and Y. Zhang, *Revisiting supernova constraints on dark gauge bosons*, arXiv:2610.35379.

[28] J. P. Leveille, *The Second Order Weak Correction to (G-2) of the Muon in Arbitrary Gauge Models*, *Nucl. Phys. B* **137** (1978) 63–76.

[29] **Muon g-2** Collaboration, D. P. Aguillard et al., *Measurement of the Positive Muon Anomalous Magnetic Moment to 127 ppb*, *Phys. Rev. Lett.* **135** (2025), no. 10 101802, [arXiv:2506.03069].

[30] R. Aliberti et al., *The anomalous magnetic moment of the muon in the Standard Model: an update*, *Phys. Rept.* **1143** (2025) 1–158, [arXiv:2505.21476].

[31] J. Heeck and W. Rodejohann, *Gauged $L_\mu - L_\tau$ Symmetry at the Electroweak Scale*, *Phys. Rev. D* **84** (2011) 075007, [arXiv:1107.5238].

[32] D. Borah, A. Dasgupta, and D. Mahanta, *TeV Scale Resonant Leptogenesis with $L_\mu - L_\tau$ Gauge Symmetry in the Light of Muon $(g-2)$*, *Phys. Rev. D* **104** (2021) 075006, [arXiv:2106.14410].

[33] R.-Y. He, J.-Q. Huang, J.-Y. Xu, F.-X. Yang, Z.-L. Han, and F.-L. Shao, *Heavy neutral leptons in gauged $U(1)_{L_\mu - L_\tau}$ at muon collider*, *Chin. Phys. C* **48** (2024) 093102, [arXiv:2401.14687].

[34] K. Asai, K. Hamaguchi, and N. Nagata, *Predictions for the neutrino parameters in the minimal gauged $U(1)_{L_\mu - L_\tau}$ model*, *Eur. Phys. J. C* **77** (2017), no. 11 763, [arXiv:1705.00419].

[35] **NA64** Collaboration, Y. M. Andreev et al., *Search for Light Dark Matter with NA64 at CERN*, *Phys. Rev. Lett.* **131** (2023), no. 16 161801, [arXiv:2307.02404].

[36] T. Hapitas, D. Tuckler, and Y. Zhang, *General Kinetic Mixing in Gauged $U(1)_{L_\mu - L_\tau}$ Model for Muon $g-2$ and Dark Matter*, *Phys. Rev. D* **105** (2022) 016014, [arXiv:2108.12440].

[37] Y. Kahn, G. Krnjaic, S. Mishra-Sharma, and T. M. P. Tait, *Light Weakly Coupled Axial Forces: Models, Constraints, and Projections*, *JHEP* **05** (2017) 002, [arXiv:1609.09072].

[38] R. Alonso, D. Blas, and P. Wolf, *Exploring the ultra-light to sub-mev dark matter window with atomic clocks and co-magnetometers*, *JHEP* **07** (2019) 069, [arXiv:1810.00889].

[39] S. Gori, S. Knapen, T. Lin, P. Munbodh, and B. Suter, *Spin-dependent scattering of sub-gev dark matter: Models and constraints*, *Phys. Rev. D* **112** (2025) 075019, [arXiv:2506.11191].

[40] Y.-S. Liu and G. A. Miller, *Validity of the Weizsäcker-Williams approximation and the analysis of beam dump experiments: Production of an axion, a dark photon, or a new axial-vector boson*, *Phys. Rev. D* **96** (2017), no. 1 016004, [arXiv:1705.01633].

[41] R. Z. Ferreira, M. C. D. Marsh, and E. Ravensburg, *Alp couplings to muons and electrons: A comprehensive analysis of supernova bounds*, *Phys. Dark Univ.* **53** (2026) 102435, [arXiv:2510.14469].

[42] R. Bollig, W. DeRocco, P. W. Graham, and H.-T. Janka, *Muons in supernovae: Implications for the axion-muon coupling*, *Phys. Rev. Lett.* **125** (2020) 051104, [arXiv:2005.07141]. Numerical bound and rate conventions are taken from the consolidated corrected arXiv v7.

[43] R. Bollig, W. DeRocco, P. W. Graham, and H.-T. Janka, *Erratum: Muons in supernovae: Implications for the axion-muon coupling*, *Phys. Rev. Lett.* **126** (2021) 189901.

[44] **ALEPH, DELPHI, L3, OPAL, SLD, LEP Electroweak Working Group, SLD Electroweak Group, SLD Heavy Flavour Group** Collaboration, S. Schael et al., *Precision electroweak measurements on the Z resonance*, *Phys. Rept.* **427** (2006) 257–454, [hep-ex/0509008].

[45] A. Crivellin, F. Kirk, C. A. Manzari, and M. Montull, *Global Electroweak Fit and Vector-Like Leptons in Light of the Cabibbo Angle Anomaly*, *JHEP* **12** (2020) 166, [arXiv:2008.01113].

[46] Z. Poh and S. Raby, *Vector-like Leptons: Muon g-2 Anomaly, Lepton Flavor Violation, Higgs Decays, and Lepton Non-Universality*, *Phys. Rev. D* **96** (2017), no. 1 015032, [arXiv:1705.07007].

[47] G.-y. Huang, F. S. Queiroz, and W. Rodejohann, *Gauged $L_\mu - L_\tau$ at a muon collider*, *Phys. Rev. D* **103** (2021), no. 9 095005, [arXiv:2101.04956].